

# Reti e Misure – I

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Social Computing – Lezioni 4, 5

10, 13 Ottobre 2025

# Annuncio

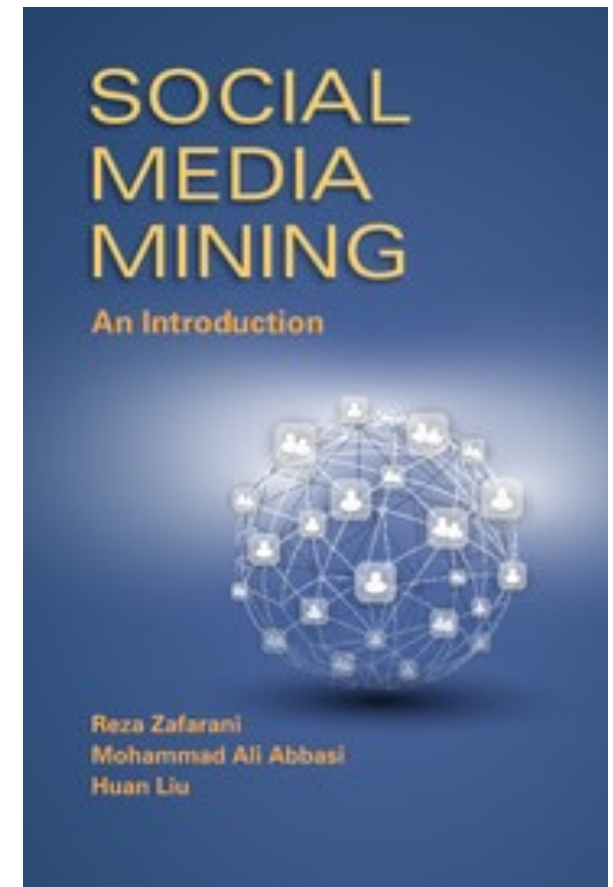
- No lezione Lun 27/10

# Riassunto

- 1. Intro a Social Computing
- 2. Intro a Social Media
  - Termini, Definizioni, Esempi, Classificazioni, Storia, Statistiche
- 3. Grafi e Reti
  - Richiami sui grafi
  - Reti = Grafi grandi nel mondo reale

# Oggi

- Misure delle reti
  - "Data una rete, che cosa possiamo misurare"
- Reza Zafarani, Mohammad Ali Abbasi, Huan Liu. *Social Media Mining - An Introduction*, Cambridge University Press, 2014
- Cap. 3 "Network Measures"
- <http://socialmediamining.info/>



# Motivazioni

- Perché misure?
- Voglio sapere, **quantificando**:
  - chi è "importante", "influyente" in una rete sociale
    - Io ho 100 follower, Obama ne ha 100 milioni...
    - Tutto qui?
    - No...
  - quali sono i pattern di interazione più tipici fra "amici"
  - (quali sono gli utenti più simili a me)
  - ...

# Esempi

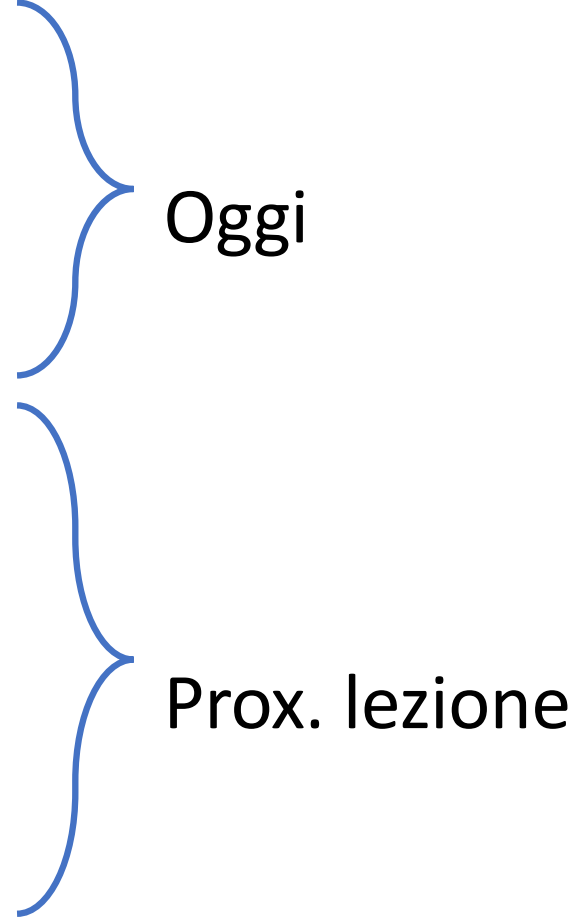
- A quali domande rispondono le misure?
- Chi sono gli individui / attori più importanti in una rete?
  - Centralità
- Quali pattern di interazione sono comuni fra amici?
  - Reciprocità e Transitività
    - Es., su Twitter / X:
      - 1. Se A è un follower di B, quanto spesso B è un follower di A?
      - 2. Se A è follower di B e B è follower di C, quanto spesso A è follower di C?
  - Bilanciamento e Status
- Quali individui sono simili e come li trovo?
  - Similitudine
- Per rispondere a queste ed altre domande, si definiscono misure **quantitative**
  - Un numero ci dà una risposta
  - Es., se per 1. si ha il 90%, si potrebbe rendere automaticamente reciproco...
  - Altrimenti, con reti grandi, sarebbe difficile capirci qualcosa

## ✦ AI Overview

The most commonly cited Galileo quote related to measurement is **"Measure what is measurable, and make measurable what is not so"**. However, this exact phrase is not found in Galileo's primary published works and may be an attribution that has emerged over time. The quote reflects his scientific approach, emphasizing the importance of quantification and the development of methods to measure phenomena, even those that are not immediately obvious or easy to quantify. 

# Scaletta

- Centralità
  - Degree
  - Betweenness
  - Closeness
  - (Ricorsiva) PageRank e HITS
- Social Network Analysis (SNA)
- Transitività e Reciprocità
  - E coefficiente di clustering
- Bilanciamento
- Distribuzione dei gradi
- Distanza media / piccolo mondo



Oggi

Prox. lezione

# Scaletta

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Oggi

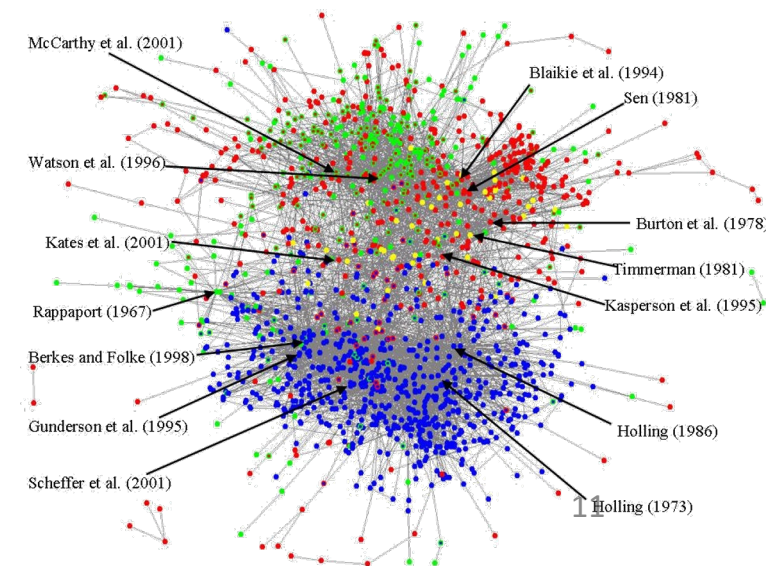
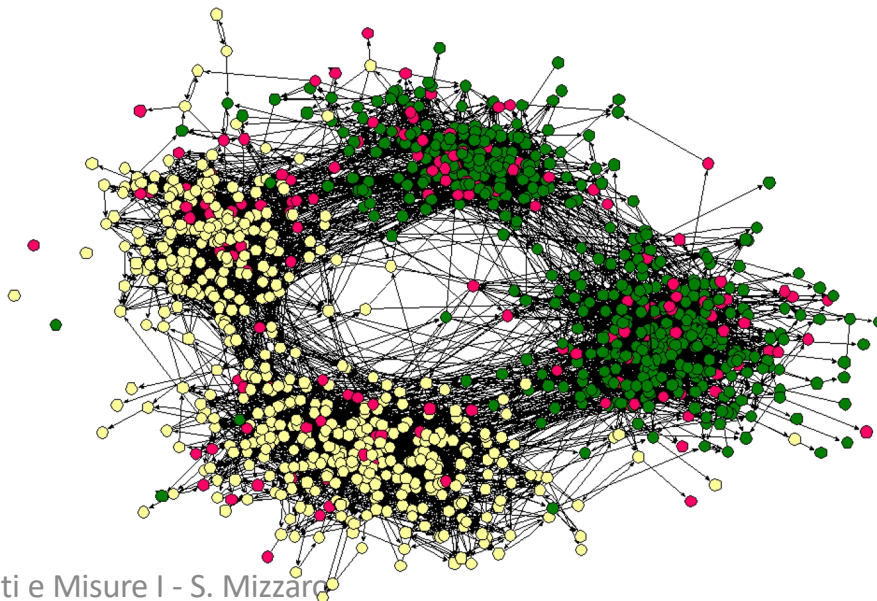
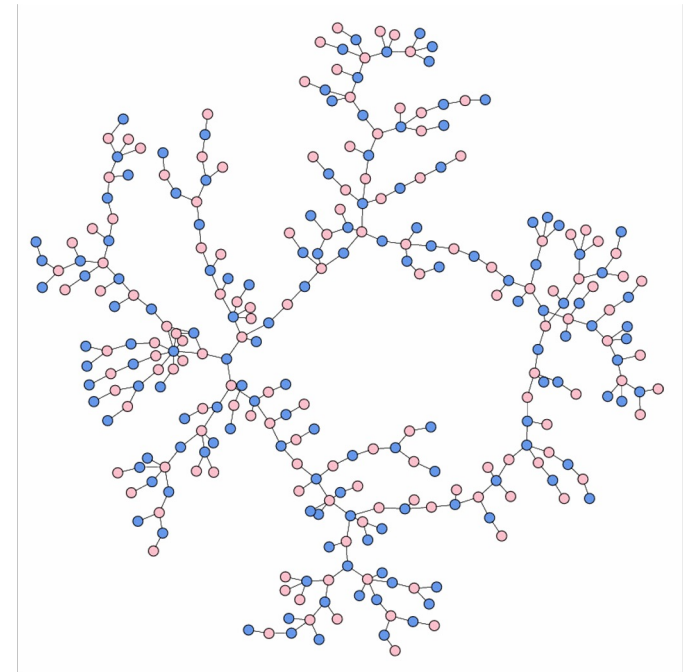
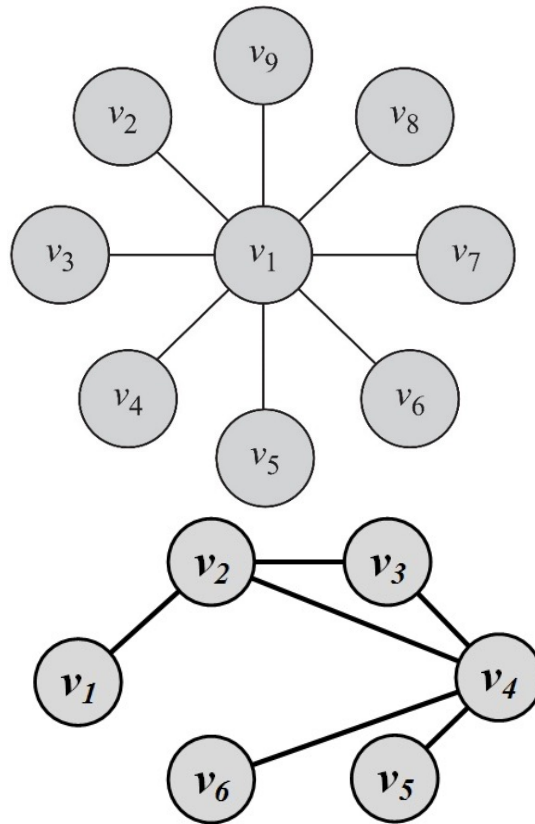
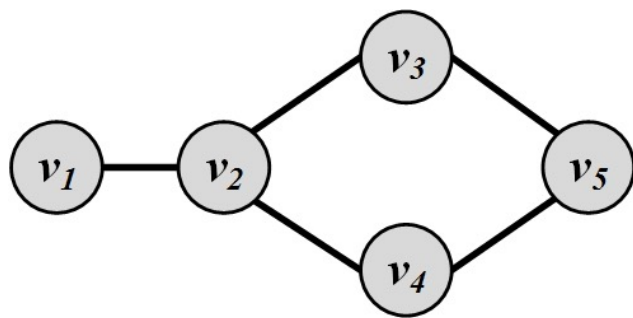


Prox. lezione

# Scaletta

- Centralità
  - Degree
  - Betweenness
  - Closeness
  - (Ricorsiva) PageRank e HITS

# Centralità



# Centralità

- "Intuitivamente"
- "Graficamente", disegno il grafo
- Il layout può trarre in inganno...
- In certi casi è facile, su reti "vere" diventa difficile/impossibile
- Quantifichiamo
- Centralità = **misura (numero)** che indica quanto un nodo è centrale
  - Sulla base di alcune definizioni/assunzioni
- Con il numero, è facile dire quale nodo è centrale!

# Klout

- Servizio che offriva analisi statistiche personalizzate sui social media
- Chiuso il 25/5/2018

|                                                |                                         |                           |
|------------------------------------------------|-----------------------------------------|---------------------------|
| <b>clout</b> <i>n</i>                          | <i>figurative, informal</i> (influence) | influenza <i>nf</i>       |
|                                                |                                         | prestigio, peso <i>nm</i> |
| The director's opinion carries a lot of clout. |                                         |                           |
| L'opinione del regista è di grande influenza.  |                                         |                           |

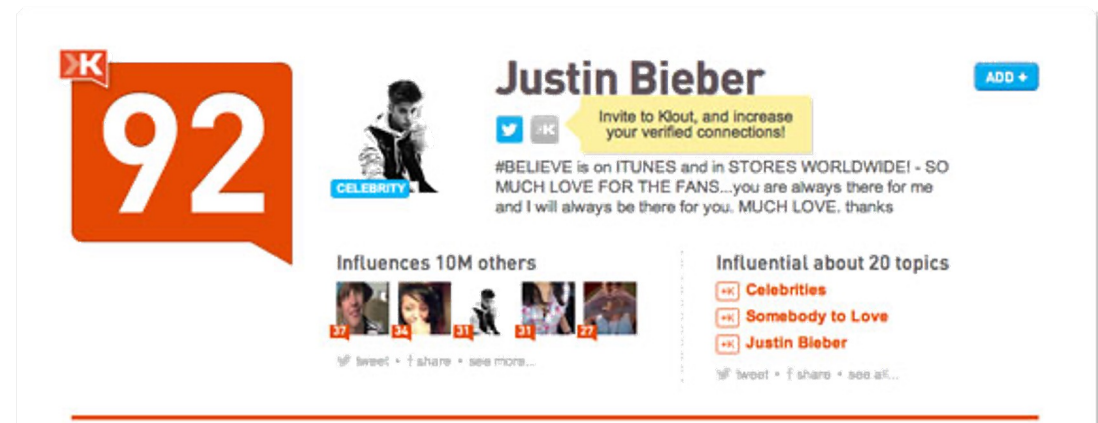
# Klout

- Servizio che offriva analisi statistiche personalizzate sui social media
- Chiuso il 25/5/2018

It is difficult  
to measure  
influence!



Klout profile for Barack Obama. The profile shows a Klout score of 99, a 'CELEBRITY' badge, and a bio stating the account is run by #Obama2012 campaign staff. It lists 'Influences 2M others' and 'Influential about 20 topics' including Government, Politics, and Media.



Klout profile for Justin Bieber. The profile shows a Klout score of 92, a 'CELEBRITY' badge, and a bio mentioning his music and fan base. It lists 'Influences 10M others' and 'Influential about 20 topics' including Celebrities, Somebody to Love, and Justin Bieber.

**KLOUT** *the Standard for Influence*



Klout Summary for Warren Buffett. The summary shows a Klout score of 36, a bio identifying him as an investor and philanthropist from Omaha, Nebraska, and a 'Score Analysis' section.

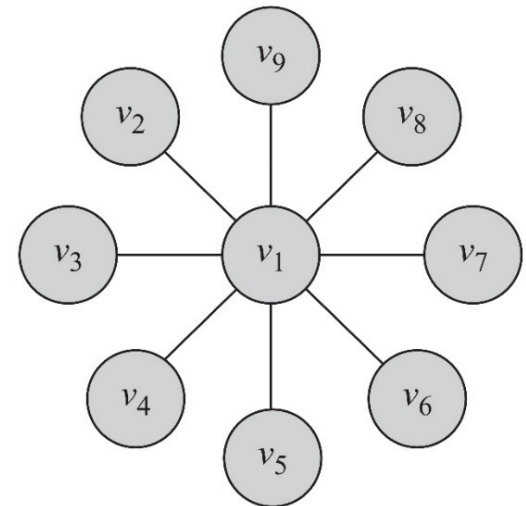
# Centralità di Grado (Degree centrality)

- "Più è alto il grado di un nodo, più è centrale"

$$C_d(v_i) = d_i$$

- $d_i$  è il grado (numero di amici) del nodo  $v_i$

- Es.:
  - Degree centrality di  $v_1$  è  $d_1=8$
  - Per gli altri  $d_j = 1, j \neq 1$



# Centralità di grado nei grafi diretti

- Possiamo usare l'in-degree, l'out-degree, e/o la combinazione dei due

$$C_d(v_i) = d_i^{\text{in}} \quad (\textit{prestige})$$

$$C_d(v_i) = d_i^{\text{out}} \quad (\textit{gregariousness})$$

$$C_d(v_i) = d_i^{\text{in}} + d_i^{\text{out}}$$

- $d_i^{\text{in}}$ : numero di archi entranti nel nodo  $v_i$
- $d_i^{\text{out}}$ : numero di archi uscenti dal nodo  $v_i$
- In pratica, in-degree più usato / interessante
  - (Perché?)

# Problema

- Immaginiamoci 2 nodi in due grafi differenti
  - $v1$  in  $G1$
  - $v2$  in  $G2$
- Centralità di grado di  $v1$ : 1000
- Centralità di grado di  $v2$ : 100
- Posso dire che  $v1$  è più centrale di  $v2$ ?
  - No, potrebbero esserci (o meno) altri nodi più centrali in entrambi i grafi!
- Aggiungo l'informazione che  $v1$  e  $v2$  sono i due nodi più centrali in  $G1$  e  $G2$ . Ora posso dire che  $v1$  è "più centrale" di  $v2$ ?
  - No!  $G1$  potrebbe avere 1 milione di nodi e  $G2$  solo 101!
- → Serve una normalizzazione

# Centralità di grado normalizzata

- Utile, ad es., per confrontare grafi differenti
- Varie possibilità
- 1. Normalizzazione per grado massimo possibile

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

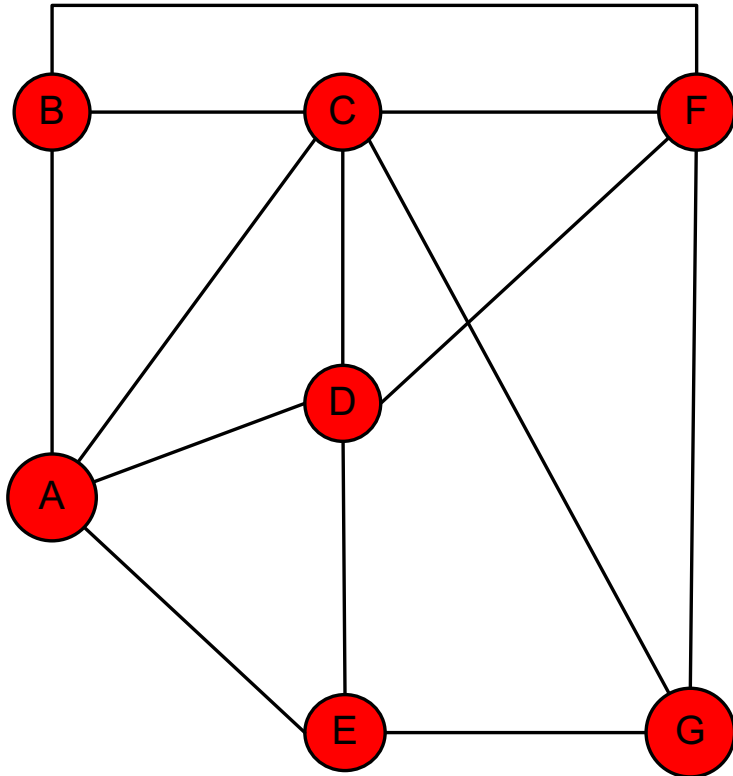
- 2. Normalizzazione per grado massimo effettivo

$$C_d^{\text{max}}(v_i) = \frac{d_i}{\max_j d_j}$$

- 3. Normalizzazione per somma dei gradi

$$C_d^{\text{sum}}(v_i) = \frac{d_i}{\sum_j d_j} = \frac{d_i}{2|E|} = \frac{d_i}{2m}$$

# Esempio (grafo indiretto)

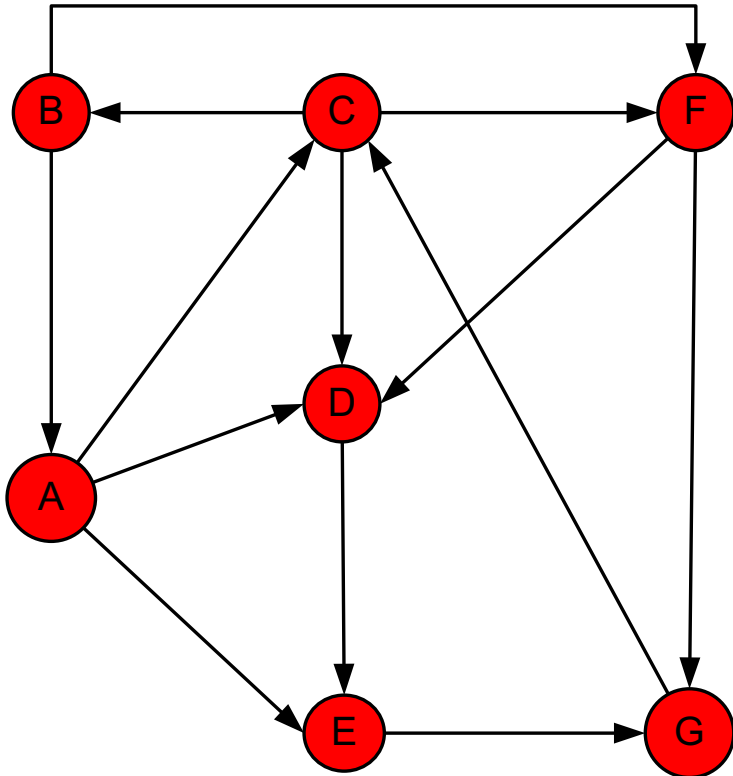


| Nodo | Grado | Centralità | Ordine |
|------|-------|------------|--------|
| A    | 4     | 2/3        | 2      |
| B    | 3     | 1/2        | 5      |
| C    | 5     | 5/6        | 1      |
| D    | 4     | 2/3        | 2      |
| E    | 3     | 1/2        | 5      |
| F    | 4     | 2/3        | 2      |
| G    | 3     | 1/2        | 5      |

1. Normalizzazione per grado massimo possibile (6)

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

# Esempio (grafo diretto, out-degree)

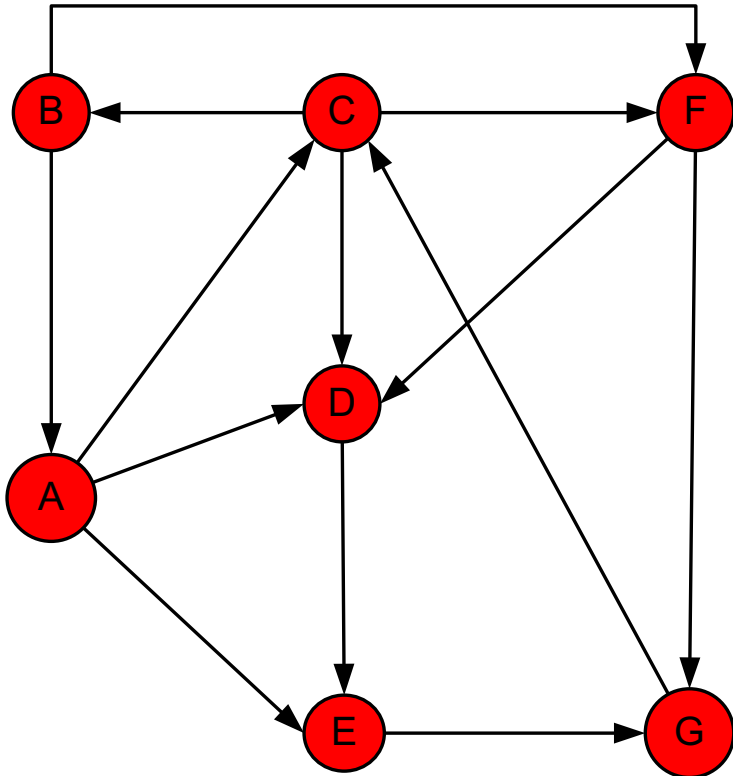


| Nodo | In-Degree | Out-Degree | Centralità | Ordine |
|------|-----------|------------|------------|--------|
| A    | 1         | 3          | 1/2        | 1      |
| B    | 1         | 2          | 1/3        | 3      |
| C    | 2         | 3          | 1/2        | 1      |
| D    | 3         | 1          | 1/6        | 5      |
| E    | 2         | 1          | 1/6        | 5      |
| F    | 2         | 2          | 1/3        | 3      |
| G    | 2         | 1          | 1/6        | 5      |

1. Normalizzazione per grado massimo possibile (6)

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

# Esempio (grafo diretto, in-degree)



| Nodo | In-Degree | Out-Degree | Centralità | Ordine |
|------|-----------|------------|------------|--------|
| A    | 1         | 3          | 1/6        | 6      |
| B    | 1         | 2          | 1/6        | 6      |
| C    | 2         | 3          | 1/3        | 2      |
| D    | 3         | 1          | 1/2        | 1      |
| E    | 2         | 1          | 1/3        | 2      |
| F    | 2         | 2          | 1/3        | 2      |
| G    | 2         | 1          | 1/3        | 2      |

1. Normalizzazione per grado massimo possibile (6)

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

# Betweenness centrality



Linton Freeman  
(Sociologo che  
propose la  
prima misura di  
betweenness  
centrality)

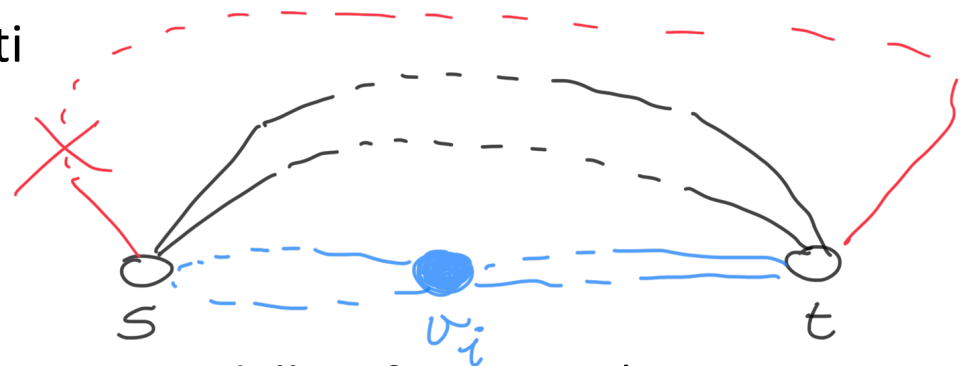
- Altro modo di guardare alla centralità: "quanto uno è in mezzo ai piedi"
- Quanto un nodo è importante nel connettere altri nodi tramite cammini
  - (si guardano solo gli shortest path)

$$C_b(v_i) = \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}}$$

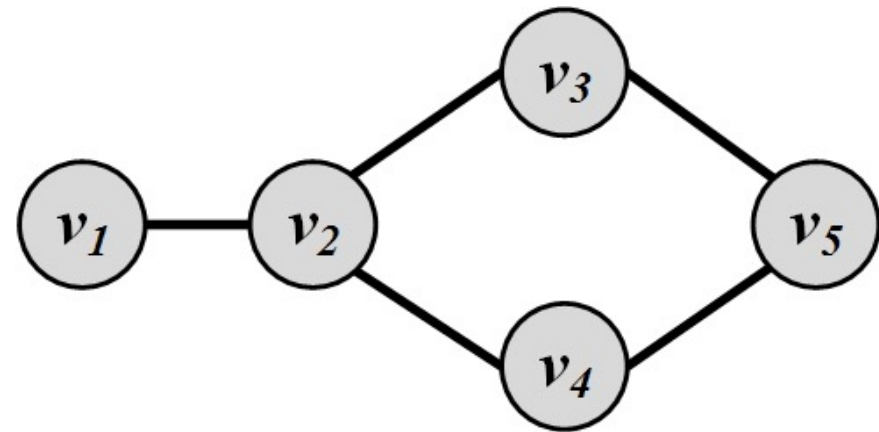
$\sigma_{st}(v_i)$  Il numero di cammini più corti (shortest path) da  $s$  a  $t$  che passano per  $v_i$

$\sigma_{st}$  Il numero di shortest path dal nodo  $s$  al nodo  $t$

(a.k.a. *information pathways, percorsi delle informazioni*)



# Esempio 1



$$C_b(v_2) = 2 \times ( \underbrace{(1/1)}_{s=v_1, t=v_3} + \underbrace{(1/1)}_{s=v_1, t=v_4} + \underbrace{(2/2)}_{s=v_1, t=v_5} + \underbrace{(1/2)}_{s=v_3, t=v_4} + \underbrace{0}_{s=v_3, t=v_5} + \underbrace{0}_{s=v_4, t=v_5} )$$

$$= 2 \times 3.5 = 7,$$

$$C_b(v_3) = 2 \times ( \underbrace{0}_{s=v_1, t=v_2} + \underbrace{0}_{s=v_1, t=v_4} + \underbrace{(1/2)}_{s=v_1, t=v_5} + \underbrace{0}_{s=v_2, t=v_4} + \underbrace{(1/2)}_{s=v_2, t=v_5} + \underbrace{0}_{s=v_4, t=v_5} )$$

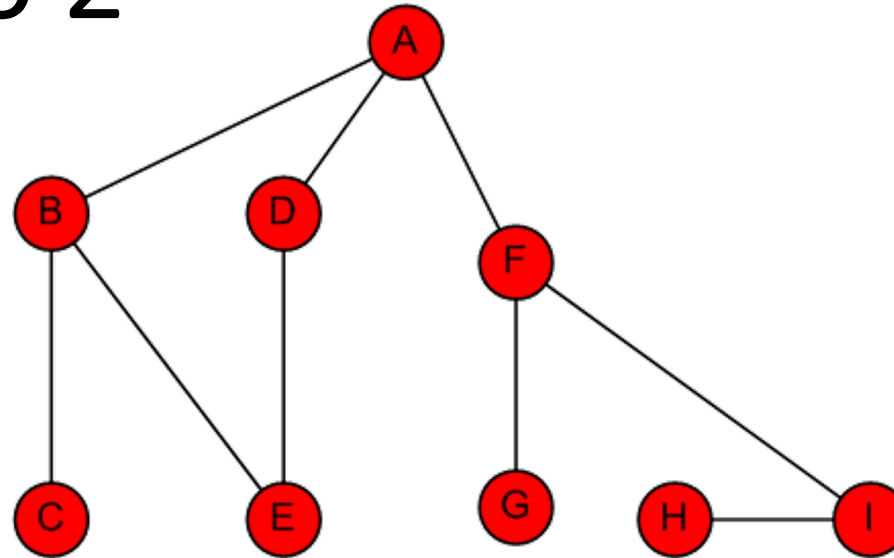
$$= 2 \times 1.0 = 2,$$

$$C_b(v_4) = C_b(v_3) = 2 \times 1.0 = 2,$$

$$C_b(v_5) = 2 \times ( \underbrace{0}_{s=v_1, t=v_2} + \underbrace{0}_{s=v_1, t=v_3} + \underbrace{0}_{s=v_1, t=v_4} + \underbrace{0}_{s=v_2, t=v_3} + \underbrace{0}_{s=v_2, t=v_4} + \underbrace{(1/2)}_{s=v_3, t=v_4} )$$

$$= 2 \times 0.5 = 1,$$

# Esempio 2



| Node | Betweenness Centrality | Rank     |
|------|------------------------|----------|
| A    | $16 + 1/2 + 1/2$       | <b>1</b> |
| B    | $7 + 5/2$              | <b>3</b> |
| C    | 0                      | <b>7</b> |
| D    | $5/2$                  | <b>5</b> |
| E    | $1/2 + 1/2$            | <b>6</b> |
| F    | $15 + 2$               | <b>1</b> |
| G    | 0                      | <b>7</b> |
| H    | 0                      | <b>7</b> |
| I    | 7                      | <b>4</b> |

# Betweenness centrality normalizzata

- Nel caso migliore, il nodo  $v_i$  è su tutti gli shortest path da  $s$  a  $t$ , quindi  $\frac{\sigma_{st}(v_i)}{\sigma_{st}} = 1, \forall s, t$

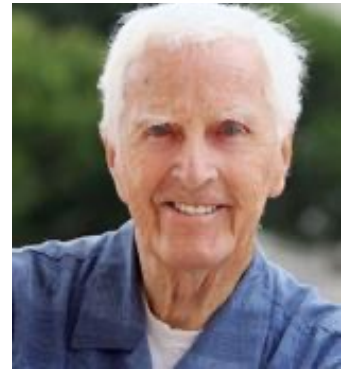
$$\begin{aligned} C_b(v_i) &= \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}} \\ &= \sum_{s \neq t \neq v_i} 1 = 2 \binom{n-1}{2} = (n-1)(n-2) \end{aligned}$$

Quindi il valore massimo è  $(n-1)(n-2)$

Normalized Betweenness centrality:

$$C_b^{\text{norm}}(v_i) = \frac{C_b(v_i)}{2 \binom{n-1}{2}}$$

# Centralità di vicinanza (Closeness centrality)



Linton Freeman

- I nodi influenti/centrali possono raggiungere velocemente gli altri nodi
  - Considero gli shortest path verso tutti gli altri nodi
  - Devono avere lunghezza media (più) bassa

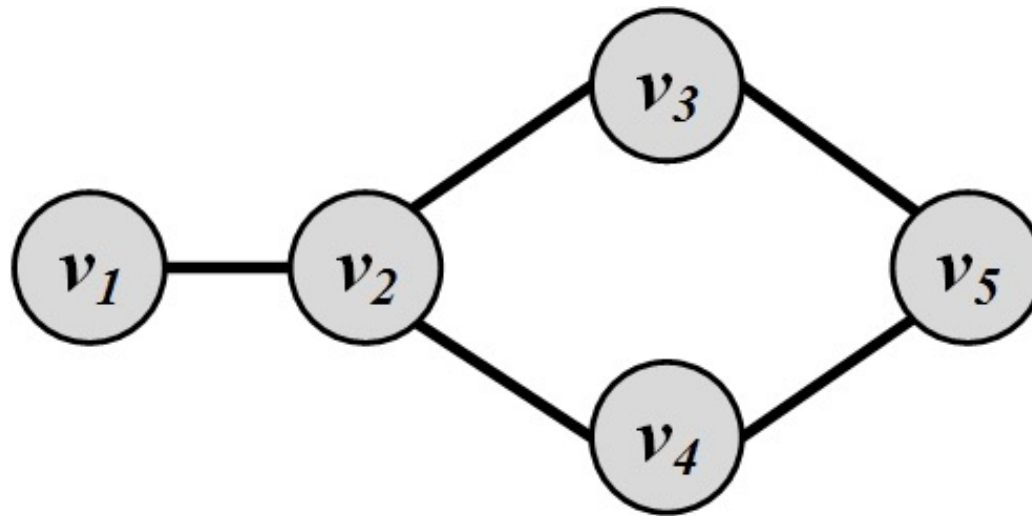
$$C_c(v_i) = \frac{1}{\bar{l}_{v_i}}$$
$$\bar{l}_{v_i} = \frac{1}{n-1} \sum_{v_j \neq v_i} l_{i,j}$$

dove

- $l_{i,j}$  : lunghezza del cammino più breve da  $v_i$  a  $v_j$
- $\bar{l}_{v_i}$  : lunghezza media dei cammini più brevi da  $v_i$

# Esempio 1

$$C_c(v_i) = \frac{1}{\bar{l}_{v_i}}$$
$$\bar{l}_{v_i} = \frac{1}{n-1} \sum_{v_j \neq v_i} l_{i,j}$$



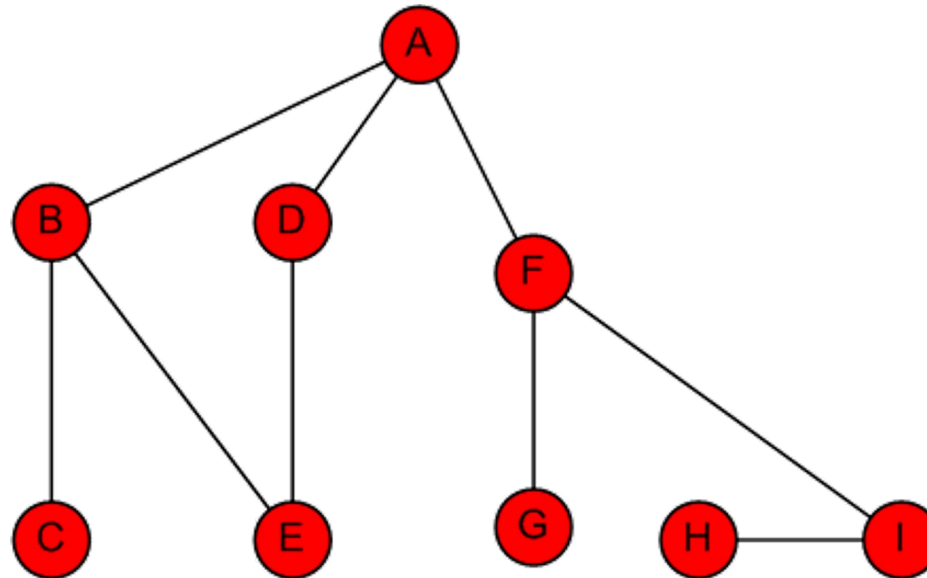
$$C_c(v_1) = 1 / ( (1 + 2 + 2 + 3)/4 ) = 0.5,$$

$$C_c(v_2) = 1 / ( (1 + 1 + 1 + 2)/4 ) = 0.8,$$

$$C_c(v_3) = C_b(v_4) = 1 / ( (1 + 1 + 2 + 2)/4 ) = 0.66,$$

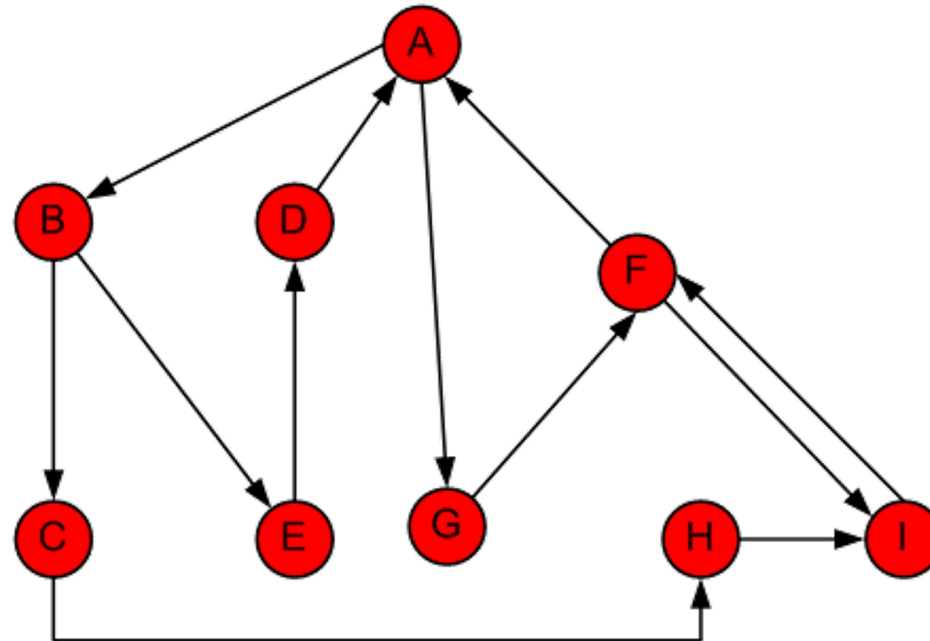
$$C_c(v_5) = 1 / ( (1 + 1 + 2 + 3)/4 ) = 0.57.$$

# Esempio 2 (indiretto)



| Node | A | B | C | D | E | F | G | H | I | D_Avg | Closeness Centrality | Rank     |
|------|---|---|---|---|---|---|---|---|---|-------|----------------------|----------|
| A    | 0 | 1 | 2 | 1 | 2 | 1 | 2 | 3 | 2 | 1.750 | 0.571                | <b>1</b> |
| B    | 1 | 0 | 1 | 2 | 1 | 2 | 3 | 4 | 3 | 2.125 | 0.471                | <b>3</b> |
| C    | 2 | 1 | 0 | 3 | 2 | 3 | 4 | 5 | 4 | 3.000 | 0.333                | <b>8</b> |
| D    | 1 | 2 | 3 | 0 | 1 | 2 | 3 | 4 | 3 | 2.375 | 0.421                | <b>4</b> |
| E    | 2 | 1 | 2 | 1 | 0 | 3 | 4 | 5 | 4 | 2.750 | 0.364                | <b>7</b> |
| F    | 1 | 2 | 3 | 2 | 3 | 0 | 1 | 2 | 1 | 1.875 | 0.533                | <b>2</b> |
| G    | 2 | 3 | 4 | 3 | 4 | 1 | 0 | 3 | 2 | 2.750 | 0.364                | <b>7</b> |
| H    | 3 | 4 | 5 | 4 | 5 | 2 | 3 | 0 | 1 | 3.375 | 0.296                | <b>9</b> |
| I    | 2 | 3 | 4 | 3 | 4 | 1 | 2 | 1 | 0 | 2.500 | 0.400                | <b>5</b> |

# Esempio 3 (diretto)



| Node | A | B | C | D | E | F | G | H | I | D_Avg | Closeness Centrality | Rank     |
|------|---|---|---|---|---|---|---|---|---|-------|----------------------|----------|
| A    | 0 | 1 | 2 | 3 | 2 | 2 | 1 | 3 | 3 | 2.125 | 0.471                | <b>1</b> |
| B    | 3 | 0 | 1 | 2 | 1 | 4 | 4 | 2 | 3 | 2.500 | 0.400                | <b>2</b> |
| C    | 4 | 5 | 0 | 7 | 6 | 3 | 5 | 1 | 2 | 4.125 | 0.242                | <b>9</b> |
| D    | 1 | 2 | 3 | 0 | 3 | 3 | 2 | 4 | 5 | 2.875 | 0.348                | <b>3</b> |
| E    | 2 | 3 | 4 | 1 | 0 | 4 | 3 | 5 | 5 | 3.375 | 0.296                | <b>6</b> |
| F    | 1 | 2 | 3 | 4 | 3 | 0 | 2 | 4 | 4 | 2.875 | 0.348                | <b>4</b> |
| G    | 2 | 3 | 4 | 5 | 4 | 1 | 0 | 5 | 2 | 3.250 | 0.308                | <b>5</b> |
| H    | 4 | 4 | 5 | 6 | 5 | 2 | 4 | 0 | 1 | 3.875 | 0.258                | <b>8</b> |
| I    | 2 | 3 | 4 | 5 | 4 | 1 | 4 | 5 | 0 | 3.500 | 0.286                | <b>7</b> |

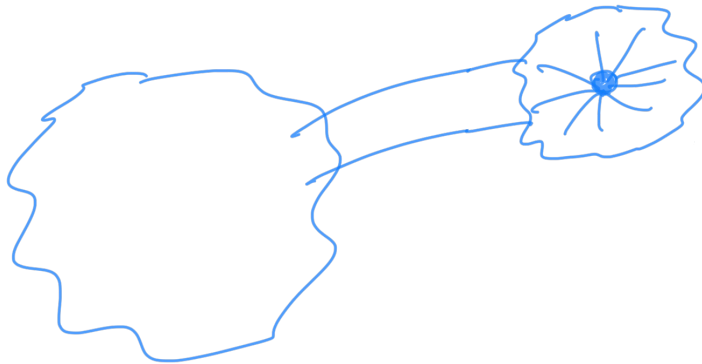
# Confronti interessanti

- Confrontando i tre valori di centralità:
  - **Di solito i tre valori hanno correlazione positiva**
  - Quando non è così (negativa o bassa), ciò dà info utili

|                     | Low<br>Degree                                        | Low<br>Closeness                                                                        | Low<br>Betweenness                                                                             |
|---------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| High<br>Degree      |                                                      | Node is embedded in a <b>community</b> that is far from the rest of the network         | Ego's connections are <b>redundant</b> - communication bypasses the node                       |
| High<br>Closeness   | <b>Key node</b> connected to important/active alters |                                                                                         | Probably <b>multiple paths</b> in the network, ego is near many people, but so are many others |
| High<br>Betweenness | <b>Ego's few ties are crucial</b> for network flow   | Very rare! Ego <b>monopolizes the ties from a small number of people to many others</b> |                                                                                                |

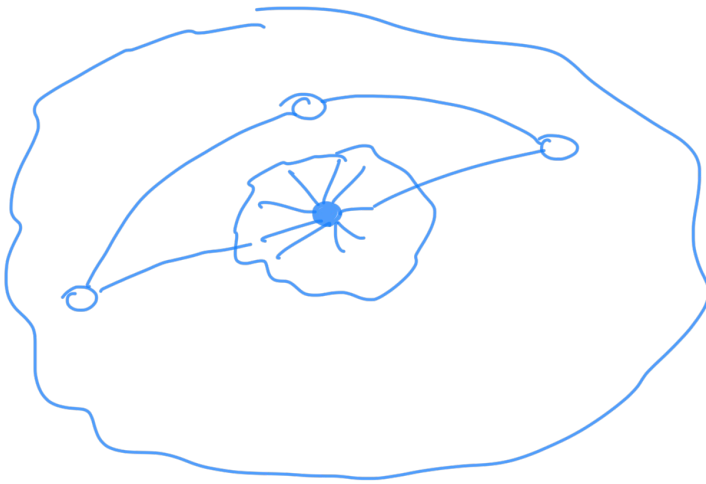
# High Degree – Low Closeness

- Node is embedded in a community that is far from the rest of the network



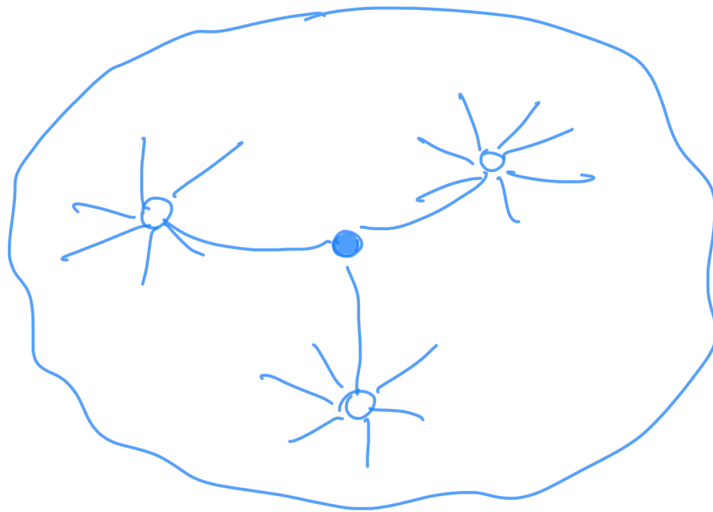
# High Degree – Low Betweenness

- Ego's connections are redundant - communication bypasses the node



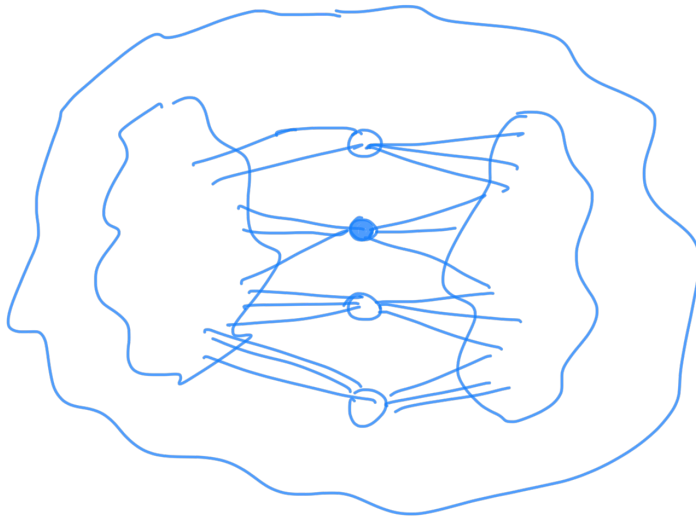
# High Closeness – Low Degree

- Key node connected to important/active alters



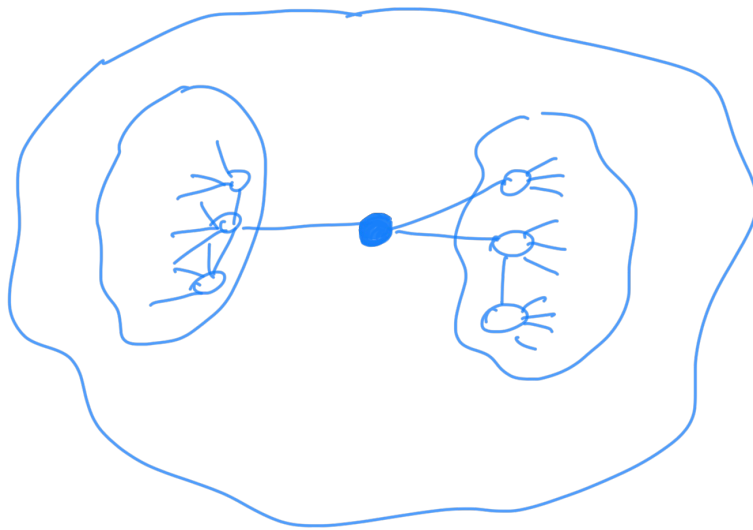
# High Closeness – Low Betweenness

- Probably multiple paths in the network, ego is near many people, but so are many others



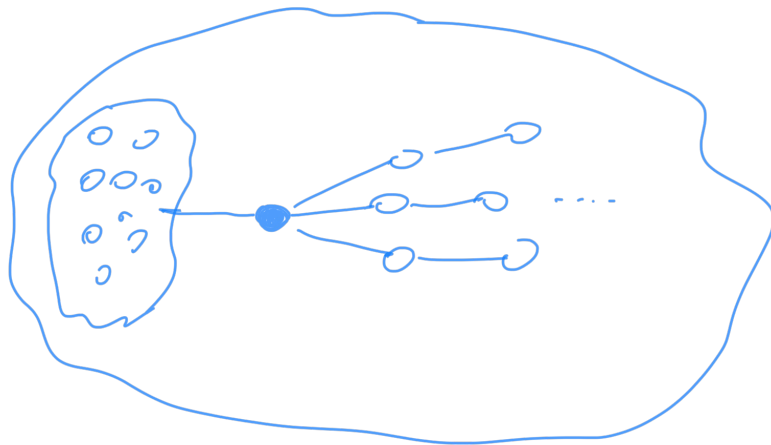
# High Betweenness – Low Degree

- Ego's few ties are crucial for network flow



# High Betweenness – Low Closeness

- Very rare! Ego monopolizes the ties from a small number of people to many others



# Scaletta

- Centralità
  - Degree
  - Betweenness
  - Closeness
  - (Ricorsiva) PageRank e HITS

# Nota

- Lucidi in inglese...
- Lucidi "per il Web"
- Concetti nati storicamente per il Grafo del Web
  - Nodi = pagine web
  - Archi = link ipertestuali
- Facile trasformarli per una rete (sociale) qualsiasi
  - (Web) Page → nodo → Profilo/utente
  - Link → arco (diretto) → Legame di amicizia, follower, ...

# Centralità di grado nei grafi diretti

- Possiamo usare l'in-degree, l'out-degree, e/o la combinazione dei due

$$C_d(v_i) = d_i^{\text{in}} \quad (\textit{prestige})$$

$$C_d(v_i) = d_i^{\text{out}} \quad (\textit{gregariousness})$$

$$C_d(v_i) = d_i^{\text{in}} + d_i^{\text{out}}$$

- $d_i^{\text{in}}$ : numero di archi entranti nel nodo  $v_i$
- $d_i^{\text{out}}$ : numero di archi uscenti dal nodo  $v_i$
- In pratica, in-degree più usato / interessante
  - (Perché?)

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- In pratica, **in-degree più usato / interessante**
  - (Perché?)

# Indegree?

- Però
  - Ci sono archi e archi
  - (In)Archi che vengono da nodi importanti dovrebbero "contare di più" di (in)archi che vengono da nodi poco importanti
  - Read that again but slowly

# Indegree?

- Però
  - Ci sono archi e archi
  - (In)Archi che vengono da **nodi importanti** dovrebbero "contare di più" di (in)archi che vengono da **nodi poco importanti**
  - Read that again but slowly

# Indegree?

- Però

- Ci sono archi e archi
- (In)Archi che vengono da **nodi importanti** dovrebbero "contare di più" di (in)archi che vengono da **nodi poco importanti**
- Read that again but slowly

**aumentare di più la centralità**

**più centrali!**

**meno centrali!**

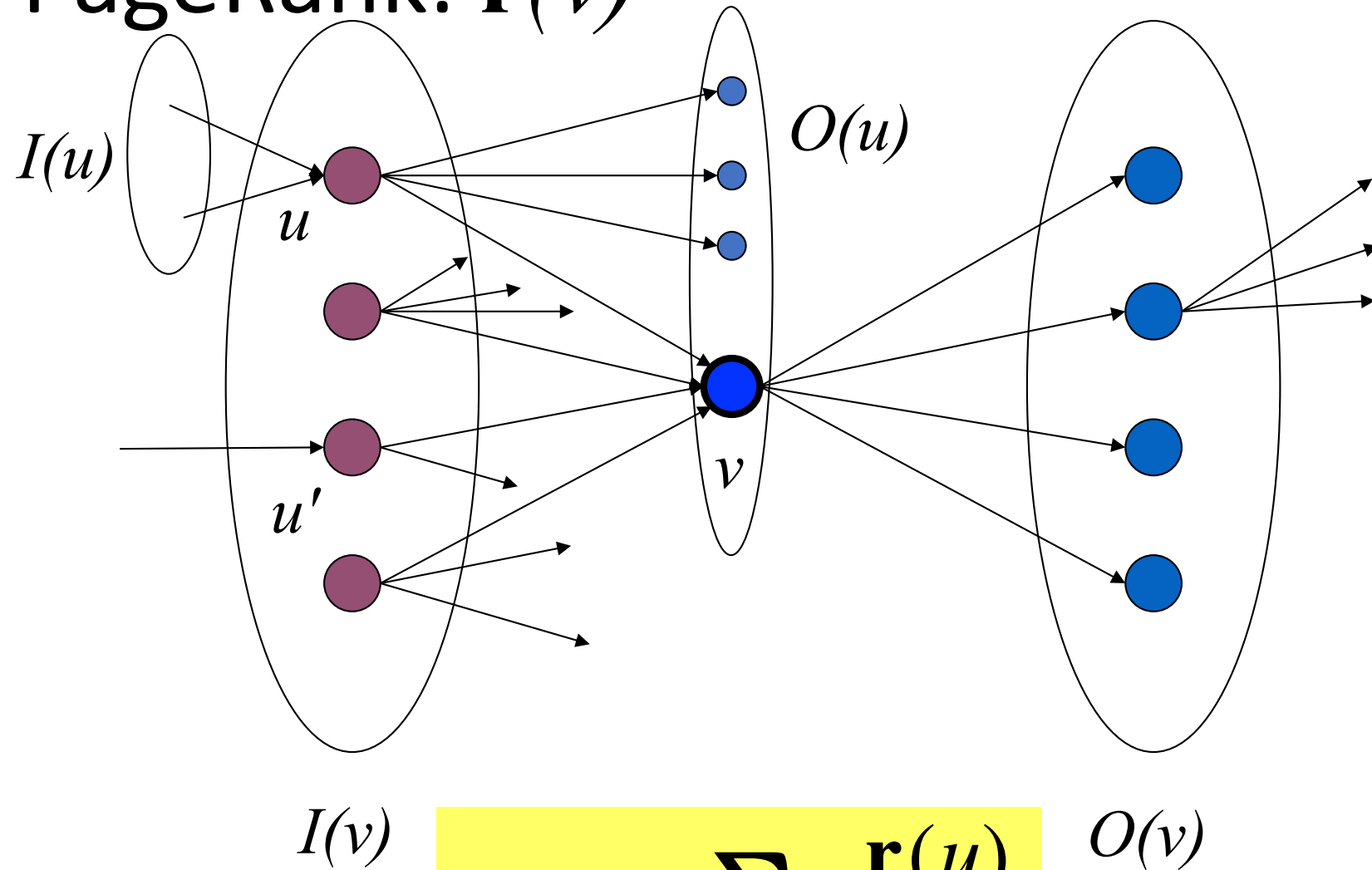
# L'idea

- La centralità di un nodo aumenta di più se riceve un arco da un nodo più centrale rispetto a ricevere un arco da un nodo meno centrale

# PageRank, "fortune of Google" (?)

- [Page et al. 98] [Brin & Page 98]
- "PageRank of page  $u$  is high if  $u$  is linked from pages with high PageRank"
  - from many pages
  - from pages having high PageRank
  - both: from many high PageRank pages
- (Query independent: each web page has its own "quality" score)
- PageRank name: from web page; from Larry Page
- Let's be more precise

# PageRank: $\mathbf{r}(v)$



$$\mathbf{r}(v) \approx \sum_{u \in I(v)} \frac{\mathbf{r}(u)}{|O(u)|}$$

# In words (1/2)

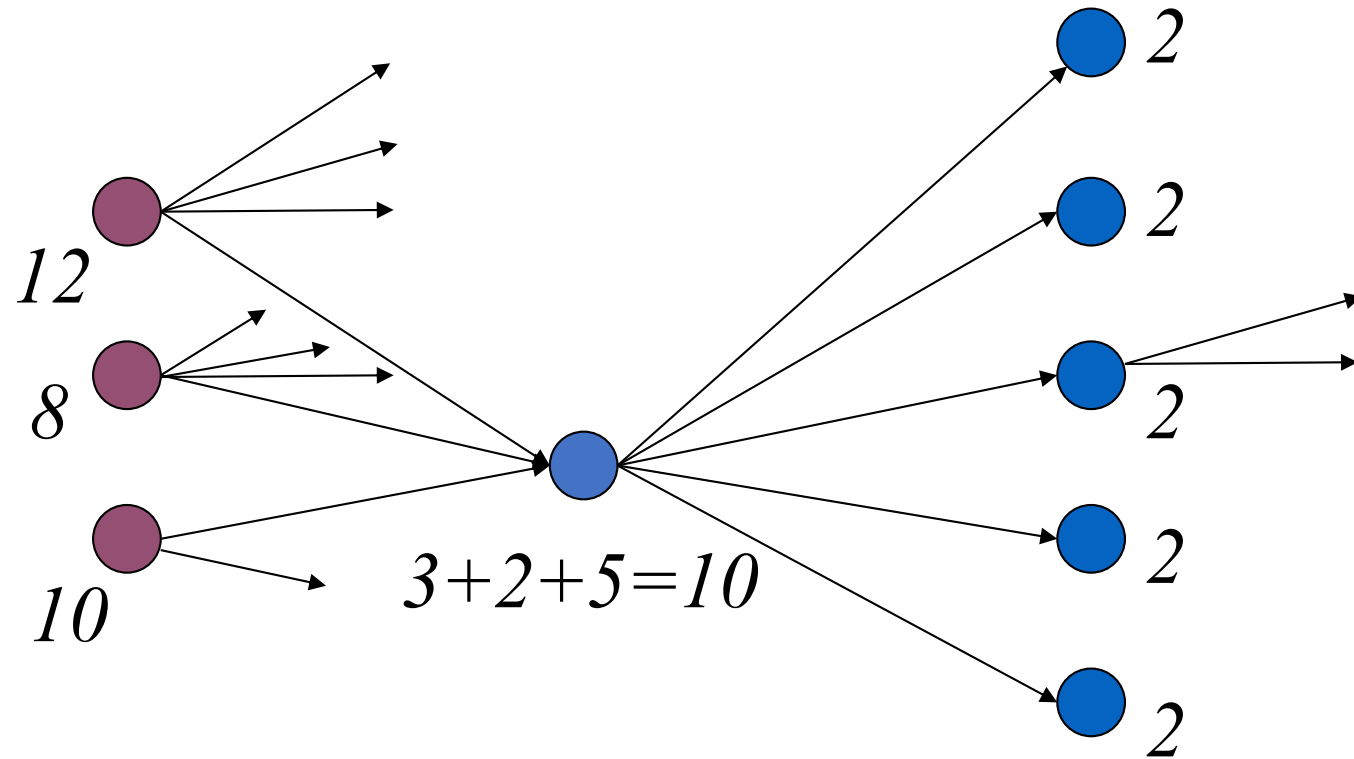
- Notation:
  - $\mathbf{r}(v)$  PageRank of page  $v$
  - $I(v)$  set of pages that link  $v$
  - $O(v)$  set of pages that are linked from  $v$
- Each page  $u$  has its PageRank value  $\mathbf{r}(u)$
- Page  $v$  "receives" from page  $u$  a portion of  $u$ 's PageRank equal to the portion received by any other page linked from  $u$ :

$$\frac{\mathbf{r}(u)}{|O(u)|}$$

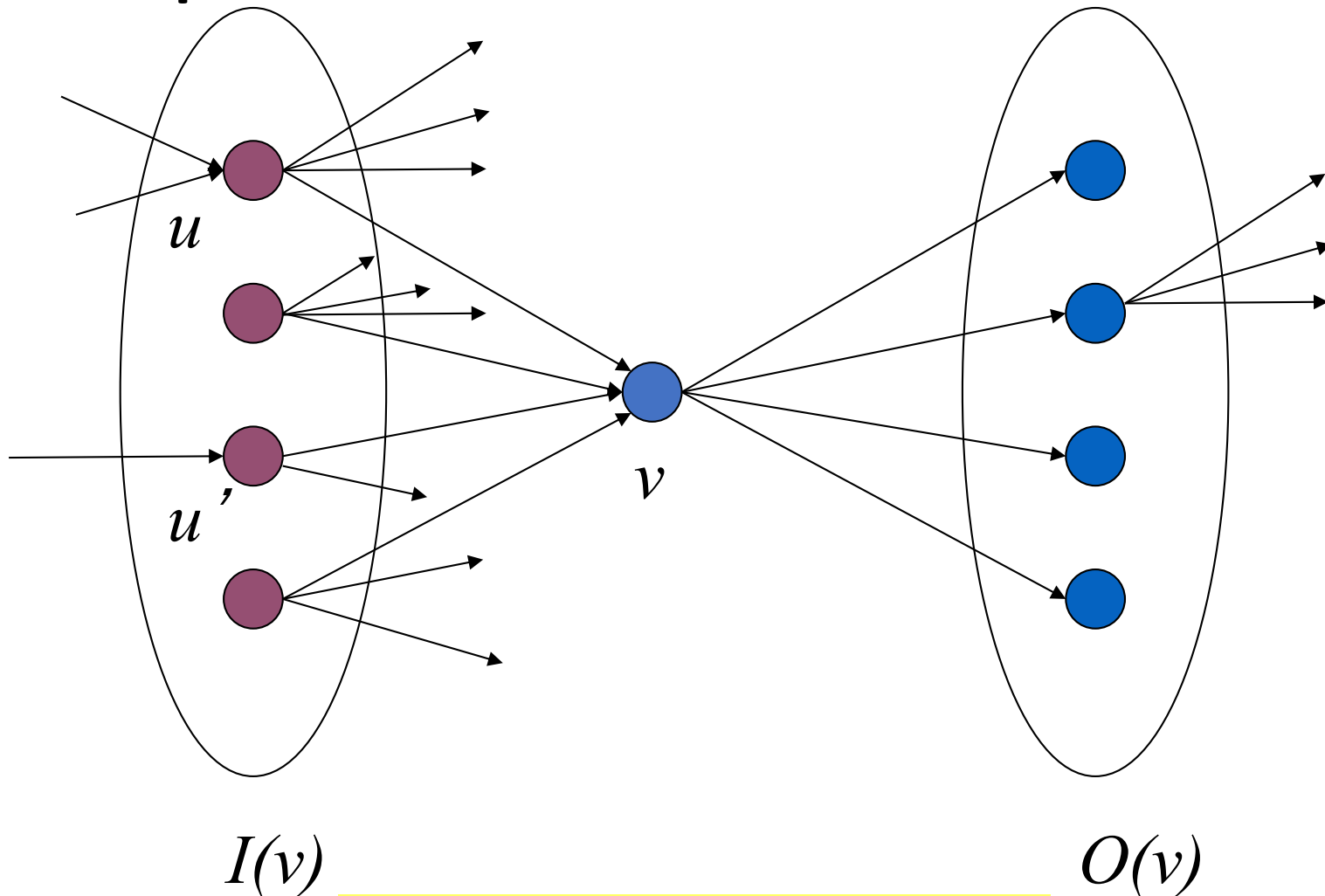
# In words (2/2)

- Each page  $u$  "distributes" its PageRank value  $\mathbf{r}(u)$  to the linked pages (i.e., those in  $O(u)$  set)
  - +  $\mathbf{r}(u)$  is high, + pages in  $O(u)$  gain
    - If  $\mathbf{r}(u) > \mathbf{r}(u')$  and  $|O(u)| = |O(u')|$ ,  $v$  gains more from  $u$  than from  $u'$
  - +  $O(u)$  is low, + pages in  $O(u)$  gain (because they don't have to "share"  $\mathbf{r}(u)$  with others)
    - If  $\mathbf{r}(u) = \mathbf{r}(u')$  and  $|O(u)| > |O(u')|$ ,  $v$  gains more from  $u'$  than from  $u$
- (actually, a normalization is needed)

# Example



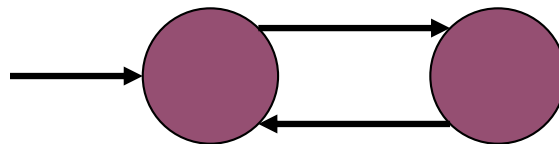
# More precise definition



$$\forall v, \mathbf{r}(v) = c \cdot \sum_{u \in I(v)} \frac{\mathbf{r}(u)}{|O(u)|}$$

# Problems (intuition)

- Initial values?
- What happens with a disconnected network?
- Loops?
- Convergence?
  - "Rank sink": it does not work...



- Before answering, let us see a formalization

# Formalization

- 1. Vectors and Matrices
- 2. Random walks
- 3. Markov chains

# Vectors and Matrices

- We can collect the PageRank values of all pages  $u_i$  in a (row) vector  $\mathbf{r}$

$$\mathbf{r} = \begin{bmatrix} \mathbf{r}(u_0) & \mathbf{r}(u_1) & \dots & \mathbf{r}(u_n) \end{bmatrix}$$

- We build/compute  $\mathbf{r}$  **iteratively, all together**
- Let's start from an initial  $\mathbf{r}_0$ : how to go to  $\mathbf{r}_1$ ,  $\mathbf{r}_2$ , etc.?
- By building a matrix  $\mathbf{P}$  such that  $\mathbf{r}_{i+1} = \mathbf{r}_i \cdot \mathbf{P}$
- Repeat until a termination condition is reached
  - $\mathbf{r}_{i+1} = \mathbf{r}_i$  or  $|\mathbf{r}_{i+1} - \mathbf{r}_i| < \varepsilon$
- Convergence...

# Schema

$$f(\mathbf{r}_0) = \mathbf{r}_1$$

$$f(\mathbf{r}_1) = \mathbf{r}_2$$

...

$$f(\mathbf{r}_i) = \mathbf{r}_{i+1}$$

...

$$f(\mathbf{r}_k) = \mathbf{r}_k$$

$$\mathbf{r}_0 \cdot \mathbf{P} = \mathbf{r}_1$$

$$\mathbf{r}_1 \cdot \mathbf{P} = \mathbf{r}_2$$

...

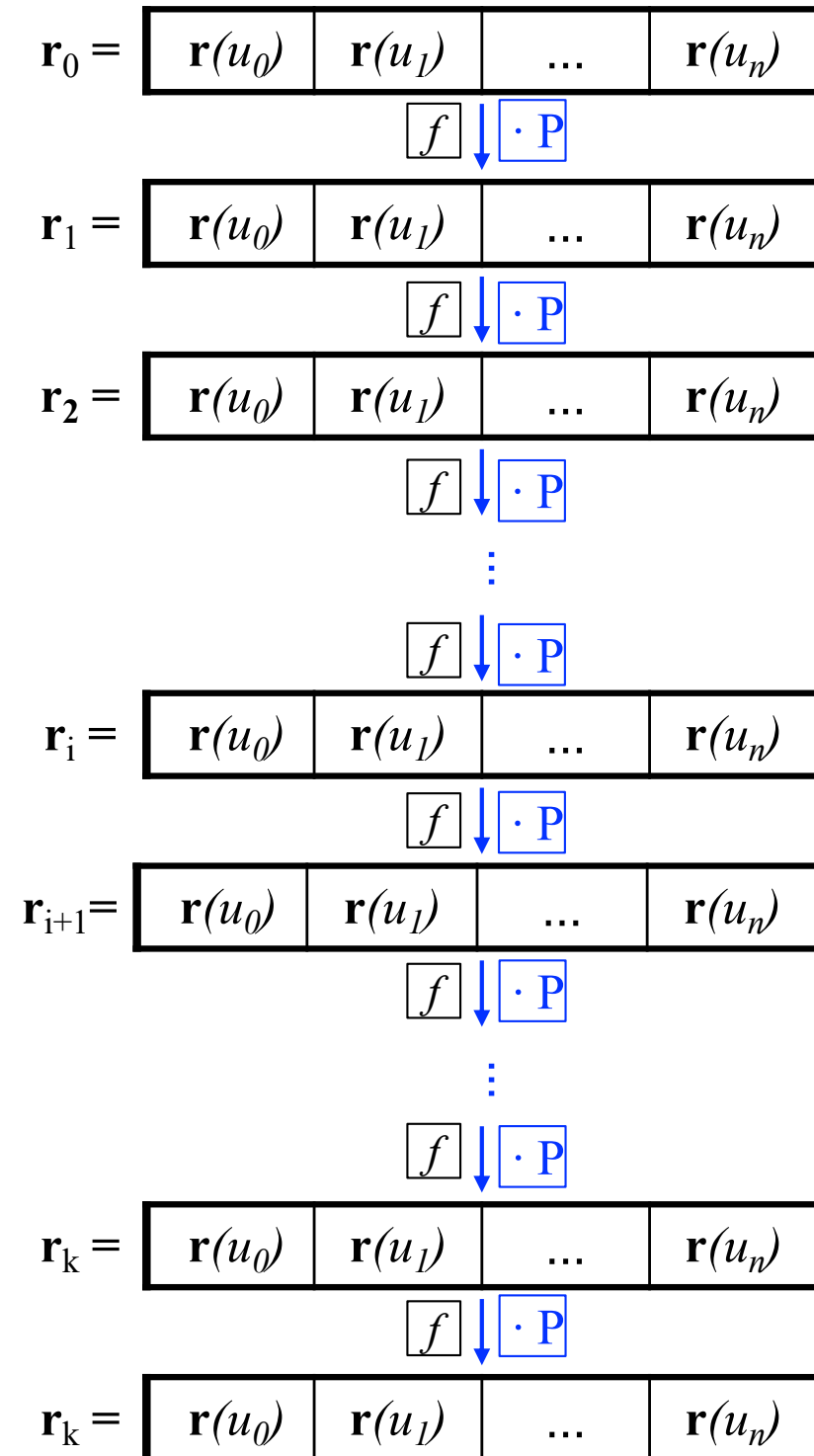
$$\mathbf{r}_i \cdot \mathbf{P} = \mathbf{r}_{i+1}$$

...

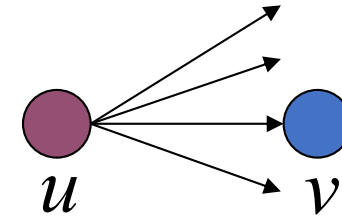
$$\mathbf{r}_k \cdot \mathbf{P} = \mathbf{r}_k$$

$$f(\mathbf{r}) = \mathbf{r}$$

$$\mathbf{r} \cdot \mathbf{P} = \mathbf{r}$$



# Random walks (1/2)



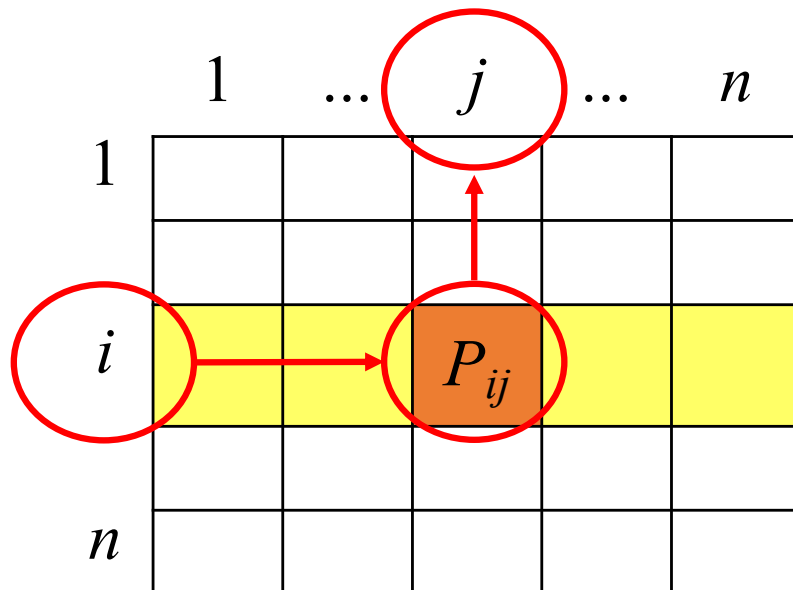
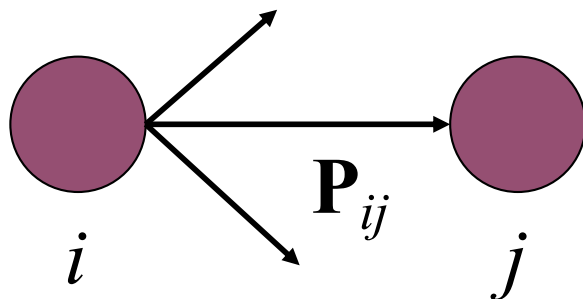
- Replace
    - "Each page distributes its PageRank value to the linked pages"
- with:
- User + Web browser
  - Let's imagine a user + browser that walks randomly among web pages
  - Starts from a random page
  - At each step/page,
    - Distributes the corresponding PageRank value
    - Selects a random link and follows it
      - All links have the same probability
  - We can replace "PageRank value of page  $v$ " with "probability that the user + browser random walk is on page  $v$ "

# Random walks (2/2)

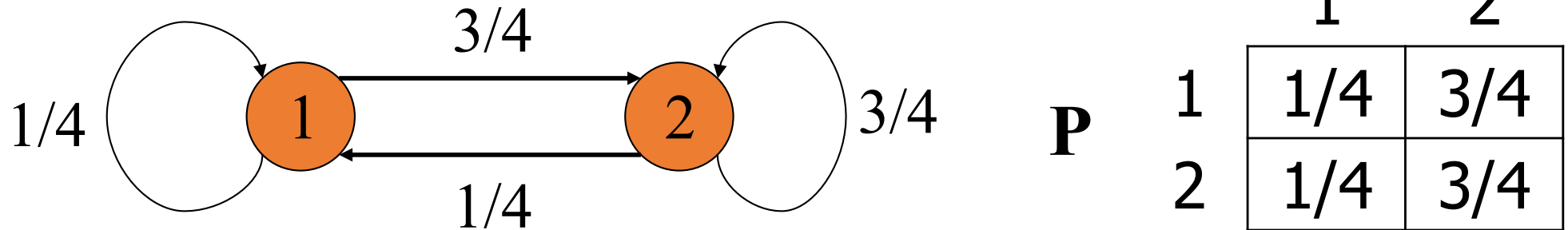
- Which is the probability that the user/walk is on a given page?
- Limit  $t \rightarrow \infty$ : if the limit exists, there is a stable state...
- PageRank of a page = probability that during a random walk the user is on that page
- Formal tool to represent random walks: Markov chain
  - Let's see what they are, then back to PageRank

# Markov chain

- Graph / (Adjacency) Matrix
- $n$  states  $\{1, 2, \dots, n\}$
- $\mathbf{P}$ : Matrix  $n \times n$  with transitions probabilities
- At a given time-step (discrete time) the Markov chain is in one of the  $n$  states
- $\forall$  current state  $i \in [1..n]$ , the Markov chain ( $\mathbf{P}$ ) tells us the probability that the next state will be  $j$ ,  $\forall j \in [1..n]$ 
  - (a step of the random walk)

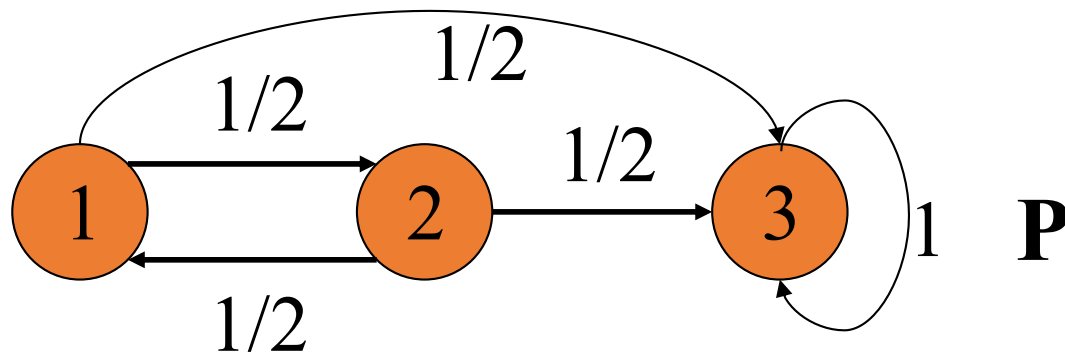


# Markov chain: An example



- Intuition: the random walk will stay more in 2 than in 1
- N.B. Sum of each row in  $\mathbf{P} = 1$  (has to go to some state, or maybe stay in the current one...)
- N.B. 2. This is even more general than "all links have the same probability"

# Markov chain: Another example



**P**

|   | 1     | 2     | 3     |
|---|-------|-------|-------|
| 1 | 0     | $1/2$ | $1/2$ |
| 2 | $1/2$ | 0     | $1/2$ |
| 3 | 0     | 0     | 1     |

- Intuition: more in 3: once there, no way to get out...

# Current position and Probability vector

- We can represent the current position in a state  $i$  by a (row) vector  $\mathbf{x} = (x_0, \dots, x_n)$ , where
  - all components have a 0 value,  $\begin{matrix} 0 & 1 & \dots & i & \dots & n \end{matrix}$
  - but the  $i$ -th, has a 1 value  $\begin{pmatrix} 0 & 0 & \dots & 1 & \dots & 0 \end{pmatrix}$
  - and representing our position in state  $i$
- Generalizing, sum of the vector values is 1
  - The Markov chain is in state  $i$  with probability  $x_i$   $\begin{matrix} 0 & 1 & \dots & i & \dots & n \\ \begin{pmatrix} x_0 & x_1 & \dots & x_i & \dots & x_n \end{pmatrix} \end{matrix}$
  - "Probability vector"

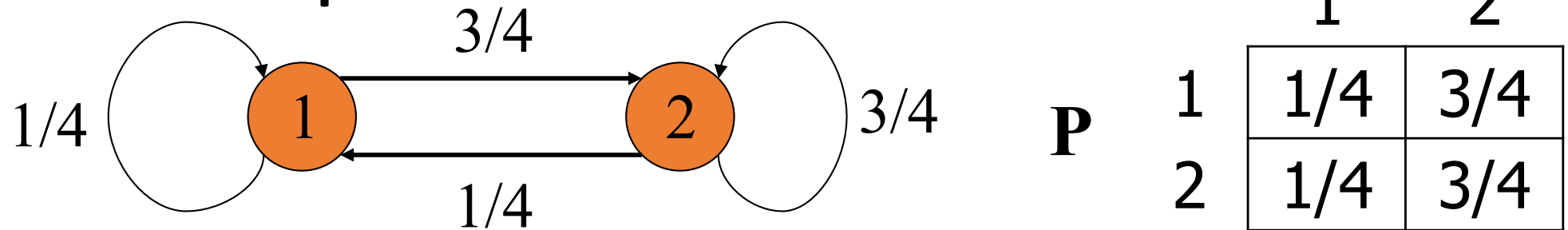
# A step of the walk

- If the probability vector is  $\mathbf{x}=(x_1, \dots, x_n)$ , what will it be at the next step of the random walk?
  - If it is like (0...010...0) (i.e., the chain is in state  $i$  with certainty)
  - Looking at the transition matrix  $\mathbf{P}$ ,  $i$ -th row, we have the probabilities to proceed to any state when starting from  $i$

$$\begin{array}{c} 1 \quad \dots \quad i \quad \dots \quad n \\ \hline \begin{array}{ccccc} \bullet & 0 & 1 & 0 & 0 \end{array} \end{array} \cdot \begin{array}{c} \begin{array}{ccccc} 1 & \dots & j & \dots & n \end{array} \\ \begin{array}{ccccc} 1 & & \bullet & & \\ & & \vdots & & \\ i & & & & \\ & & & & \\ n & & \downarrow & & \end{array} \end{array} = \begin{array}{c} 1 \quad \dots \quad i \quad \dots \quad n \\ \hline \begin{array}{ccccc} & & \bullet & & \end{array} \end{array}$$

- Generalization to values  $\neq 0/1$  is immediate (think of, e.g., two 0.5 values...)

# Example



- $\mathbf{x}_0 = (1, 0) \rightarrow \mathbf{x}_1 = (1/4, 3/4) \rightarrow \mathbf{x}_2 = (1/4, 3/4)$
- $\mathbf{x}_0 = (0, 1) \rightarrow \mathbf{x}_1 = (1/4, 3/4) \rightarrow \mathbf{x}_2 = (1/4, 3/4)$
- $\mathbf{x}_0 = (1/2, 1/2) \rightarrow \mathbf{x}_1 = (1/4, 3/4) \rightarrow \mathbf{x}_2 = (1/4, 3/4)$
- Here we have a quick convergence. This is not always the case...

# Eigenvector

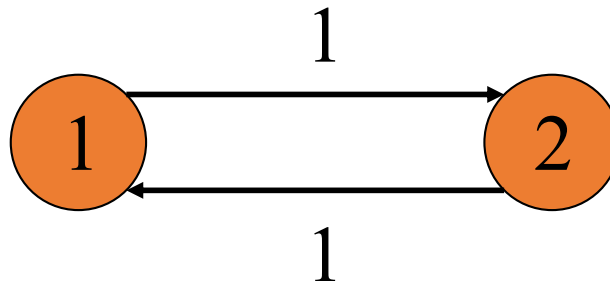
- Therefore, step of the walk:  $\mathbf{x}_{i+1} = \mathbf{x}_i \cdot \mathbf{P}$ 
  - Power iteration
- But we're looking for the stable distribution,  
 $\mathbf{x}_{i+1} = \mathbf{x}_i$
- That is, we are looking for  $\mathbf{x}$  such that  $\mathbf{x} = \mathbf{x} \cdot \mathbf{P}$ 
  - i.e., the power iteration converges
- $\mathbf{x}$  is the left eigenvector of  $\mathbf{P}$
- Does it always exist? Does the power iteration converge?

# Stable distribution

- Any **ergodic** Markov chain has a **steady state distribution**
  - Any state has a long-term visit rate. The rate converges, it is stable
  - On the long term ( $t \rightarrow \infty$ ), any state is visited proportionally to its visit rate
  - (and it is independent from the starting point of the random walk)

# Ergodic Markov chains

- There exists a path from any state to any state
  - Strongly connected
- It is possible to be in any state at any step, with probability  $> 0$
- An example of a chain connected but not ergodic:
  - (even/odd)

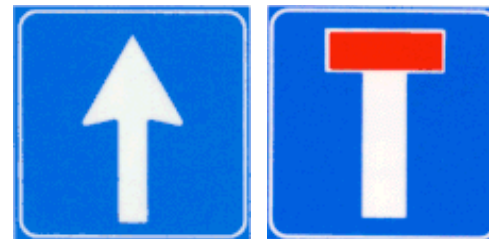
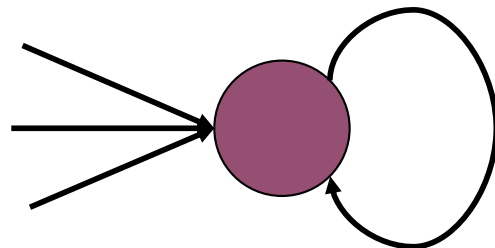


# Summary

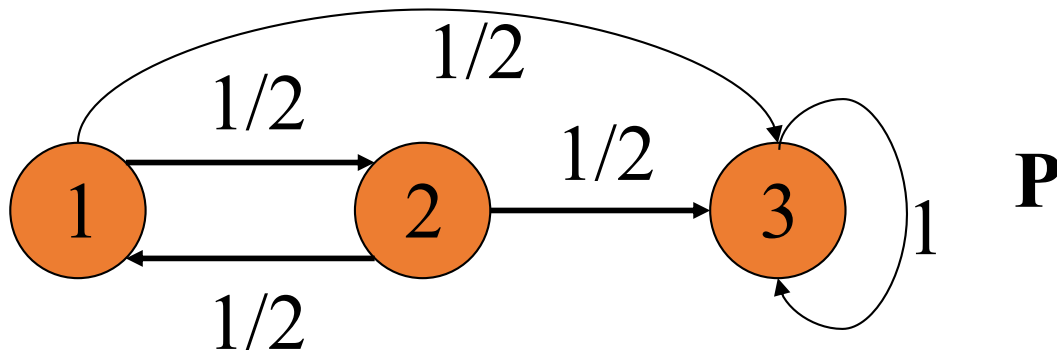
- PageRank distribution → Random walk on the Web
- We model a random walk on the Web as a Markov chain
  - Transition matrix obtained from links (Adjacency matrix)
  - Probability vector represents the position
- If it is ergodic, the probability vector has a stable limit
  - We can use that limit as PageRank
- Thus, Markov chains are a useful formal tool...
- **But: is a random walk on the Web ergodic?** 🤔
  - Of course not: the Web is not connected... 🙄
- Solution: teleporting... 🙄

# Teleporting

- During our walk on the Web, we click on a lot of links and sometimes we get bored. We then jump (we "teleport") to a random page
  - $p(\text{follow a link}) + p(\text{teleport}) = 1$
  - $p(\text{teleport}) = \text{constant} = 0.1$  (let's say)
- "Any page is linked to any other page (with low probability)"
- Also allows to exit the "cul-de-sac"s



# "Another example" revisited



**P**

|   | 1   | 2   | 3   |
|---|-----|-----|-----|
| 1 | 0   | 1/2 | 1/2 |
| 2 | 1/2 | 0   | 1/2 |
| 3 | 0   | 0   | 1   |

- Not ergodic. Let us add teleporting, with  $p(\text{teleporting}) = 0.1$

|   | 1    | 2    | 3    |
|---|------|------|------|
| 1 | 1/30 | 1/30 | 1/30 |
| 2 | 1/30 | 1/30 | 1/30 |
| 3 | 1/30 | 1/30 | 1/30 |

+

|   | 1    | 2    | 3    |
|---|------|------|------|
| 1 | 0    | 9/20 | 9/20 |
| 2 | 9/20 | 0    | 9/20 |
| 3 | 0    | 0    | 9/10 |

=

|   | 1     | 2     | 3     |
|---|-------|-------|-------|
| 1 | 1/30  | 29/60 | 29/60 |
| 2 | 29/60 | 1/30  | 29/60 |
| 3 | 1/30  | 1/30  | 14/15 |

- Now **P** is ergodic. Ex.: find the distribution
  - Trying also with several starting points

# Therefore, PageRank

- Random walk, with teleporting
- It is a Markov chain
  - Ergodic, thus it has a steady state distribution, each state has its own probability
- Any page has its own probability that during the random walk, the user is "there"
- PageRank of the pages is the probability distribution

# Formula

- Power iteration

- $\mathbf{r}_{i+1} = \mathbf{r}_i \cdot \mathbf{P}'$

$$\mathbf{r}_{i+1} = (1 - d) \cdot \mathbf{r}_i \cdot \mathbf{P} + d \cdot \mathbf{r}_i \cdot \begin{pmatrix} 1/N & \dots & 1/N \\ \vdots & \ddots & \vdots \\ 1/N & \dots & 1/N \end{pmatrix}$$

- $d$  : probability of teleporting (for ex., 0.1)
- $N$  : # of nodes/pages
- (here
  - $\mathbf{P}$  is the adjacency matrix without teleporting,
  - $\mathbf{P}'$  is the adjacency matrix with teleporting)

# Thus, PageRank computation

- Build the matrix  $\mathbf{P}'$ 
  - Adjacency matrix  $\mathbf{P}$  + teleporting (ergodic)
- Initial probability vector  $\mathbf{r}_0$ 
  - Tentative PageRank values
  - Being ergodic any  $\mathbf{r}_0$  will do, e.g.,  $1/N$ ,
  - but efficiency: re-use previous known PageRank values
- $\mathbf{r}_{i+1} = \mathbf{r}_i \cdot \mathbf{P}'$
- If  $|\mathbf{r}_{i+1} - \mathbf{r}_i| < \varepsilon$  stop, otherwise repeat
- The  $\mathbf{r}$  vector found ( $\mathbf{r} = \mathbf{r} \cdot \mathbf{P}'$ , or  $\mathbf{r} \sim \mathbf{r} \cdot \mathbf{P}'$ ) is
  - the PageRank of all pages, and  $\sim$  the left eigenvector of  $\mathbf{P}'$
- Efficiency is an issue: for the Web, the adjacency matrix  $\mathbf{P}$  has  $50B \times 50B$  elements!
  - It's very sparse ( $\sim 15$  links per page)  $\Rightarrow$  sparse matrix representation and multiplication techniques

# Pseudocode

```
pagerank(G, d = 0.15, epsilon = 0, num = +inf):  
    N = |G|  
    A = (1-d) * adjacency(G) + d * 1/N    # teleporting matrix  
    r[0] = 1/N                            # r[i] are row vectors  
    for i in [1,num[ :  
        r[i] = r[i-1] * A  
        if abs(r[i] - r[i-1]) <= epsilon:  
            return r[i]  
    return r[i]
```

# Scaletta

- Centralità
  - Degree
  - Betweenness
  - Closeness
  - (Ricorsiva)
    - PageRank e HITS

# HITS:

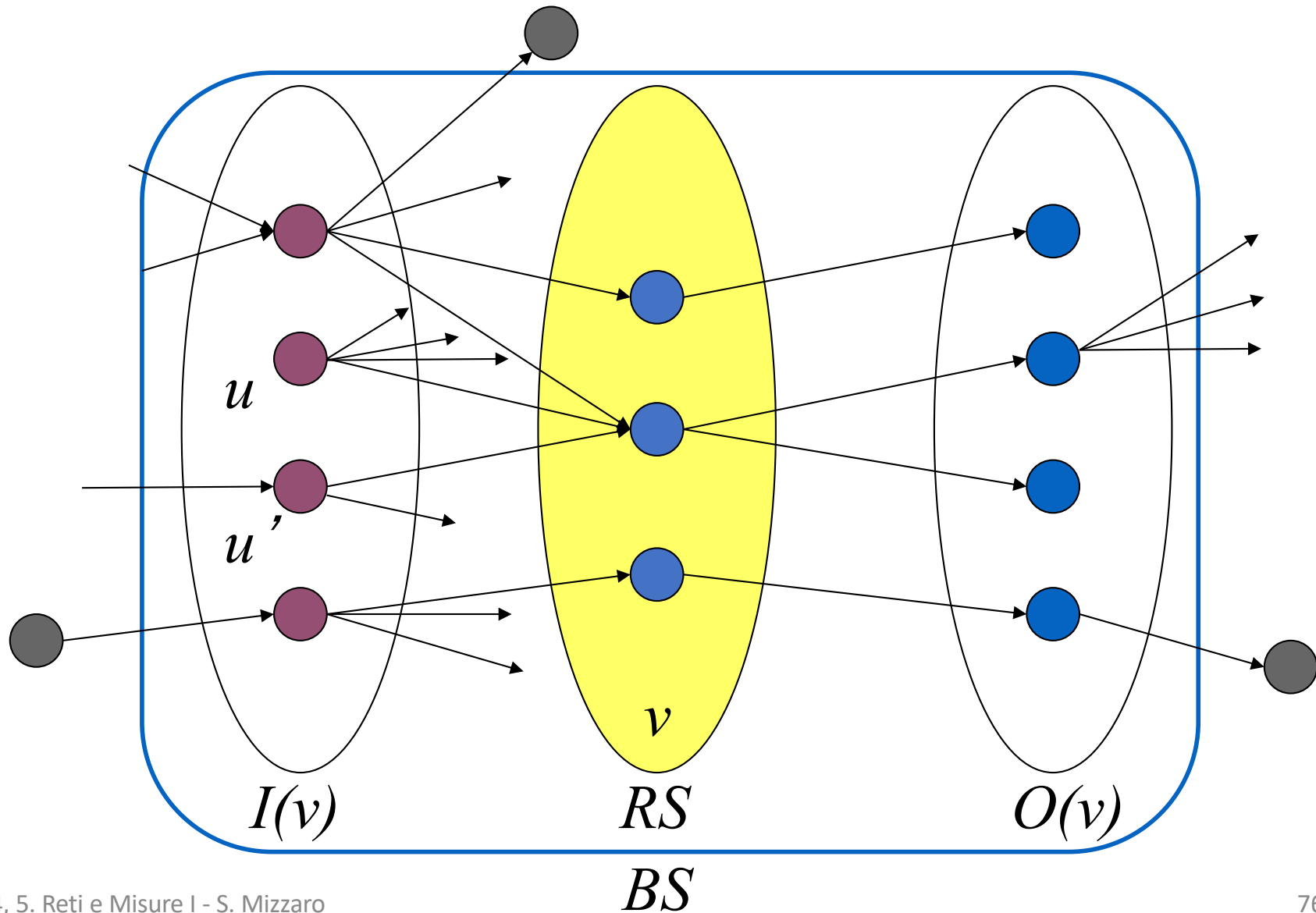
## The algorithm by Kleinberg [1999]

- "Hyperlink-Induced Topic Search"
- Idea (circular, by now. Then recursive): to build 2 related sets of pages
  - Hub pages (H) [outlinks]
    - Pages containing (good) links to other pages: yahoo, good bookmarks, ...
    - ("Hub", for airports: airports from where you can fly to many places)
  - Authority pages (A) [inlinks]
    - Pages of well recognized people/entities on the Web: w3c, Knuth's home page, ...
    - Pages that are linked a lot from other good pages
- **Good hubs link good authorities; good authorities are linked by good hubs**

# Root and Base Set

- Pages the algorithm works on:
  - Not the whole Web
  - A query time algorithm
  - The set of retrieved pages, named Root Set ( $RS$ )
- We start from  $RS$ , and we expand it with:
  - $I(v)$ : in-pages (all the pages  $u$  such that  $\exists v \in RS. u \rightarrow v$ )
  - $O(v)$ : out-pages (all the pages  $u$  such that  $\exists v \in RS. v \rightarrow u$ )
- We get the Base Set ( $BS$ )
  - Good H/A candidate pages
  - The algorithm works on the BS
- (actually, intra-host links are removed)
- H and A are selected from  $BS$  by computing, for each page  $x$  in  $BS$ ,
  - a **hubness value** ( $\mathbf{h}(x)$ ) and
  - an **authority value** ( $\mathbf{a}(x)$ )

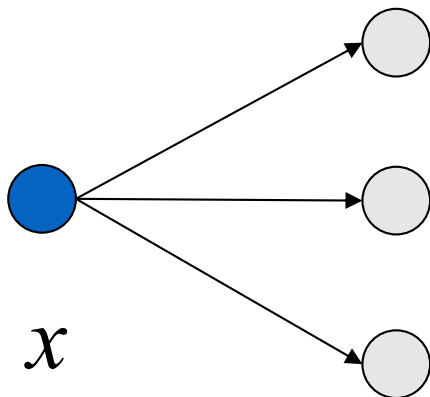
# Root Set and Base Set



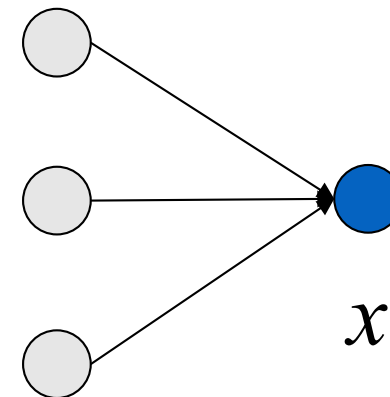
$\forall x \in BS$  compute  $\mathbf{h}(x)$  and  $\mathbf{a}(x)$

$$\mathbf{h}(x) = \sum_{x \rightarrow y} \mathbf{a}(y)$$

$$\mathbf{a}(x) = \sum_{y \rightarrow x} \mathbf{h}(y)$$



- The H value of  $x$  depends on how  $x$  links high A pages



- The A value of  $x$  depends on how  $x$  is linked from high H pages

# Procedure / "Algorithm"

- Begin by assigning initial  $\mathbf{h}(x)$  and  $\mathbf{a}(x)$  to any page  $x$  in the  $BS$ 
  - Initially  $\mathbf{h}(x) = \mathbf{a}(x) = 1$
- Then, iteratively update, with the previous formulas,  $\mathbf{h}(x)$  and  $\mathbf{a}(x)$
- Stop when converges (when  $\mathbf{h}(x)$  and  $\mathbf{a}(x)$  do not vary anymore)
- At the end, "top hub" pages and "top authority" pages

# Ex.: Qry [Japan elementary schools]

## Authorities

- The American School in Japan
- The Link Page
- %o<sup>a</sup>□è□s—§<sup>ä</sup>“c□¬Šw□Zfz□[f□fy□[fW
- Kids' Space
- ^À□é□s—§^À□é□¼•”□¬Šw□Z
- <{□é<<sup>3</sup>ç‘ăŠw•□‘®□¬Šw□Z
- KEIMEI GAKUEN Home Page ( Japanese )
- Shiranuma Home Page
- fuzoku-es.fukui-u.ac.jp
- welcome to Miasa E&J school
- □\_“p□iœ§□E%oj•l□s—  
§‘†□i□¼□¬Šw□Z,lfy
- [http://www...p/~m\\_maru/index.html](http://www...p/~m_maru/index.html)
- fukui haruyama-es HomePage
- Torisu primary school
- goo
- Yakumo Elementary,Hokkaido,Japan
- FUZOKU Home Page
- Kamishibun Elementary School...

## Hubs

- schools
- LINK Page-13
- “ú–{,lŠw□Z
- □a%o,,□¬Šw□Zfz□[f□fy□[fW
- 100 Schools Home Pages (English)
- K-12 from Japan 10/...rnet and Education )
- <http://www...iglobe.ne.jp/~IKESAN>
- ,l,f,j□¬Šw□Z,U”N,P‘g•”œê
- □ÒŠ—’¬—§□ÒŠ—“œ□¬Šw□Z
- Koulutus ja oppilaitokset
- TOYODA HOMEPAGE
- Education
- Cay's Homepage(Japanese)
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- UNIVERSITY
- %oJ—<sup>3</sup>□¬Šw□Z DRAGON97-TOP
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# Comments

- HITS works on the retrieved docs. set
  - To keep complexity low (small set)
  - To be reasonably sure that the set contains good **h** / **a**
- The iterative process does not depend on the query
  - (BS does, but once BS is fixed, the q can be forgotten)
  - Nor from the language (content independent)
- But, like Pagerank, HITS is a general algorithm
  - can be used on any graph

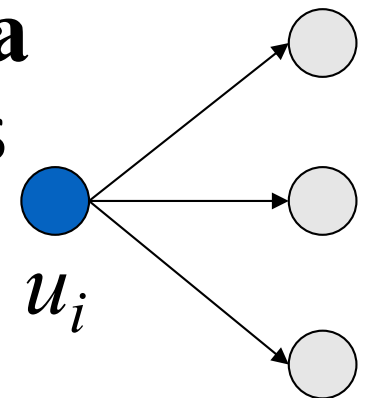
# Matrices and vectors (1/4)

- Like PageRank: matrices/vectors formalization
- Let us build 2 (column!) vectors, for **h** and **a**:

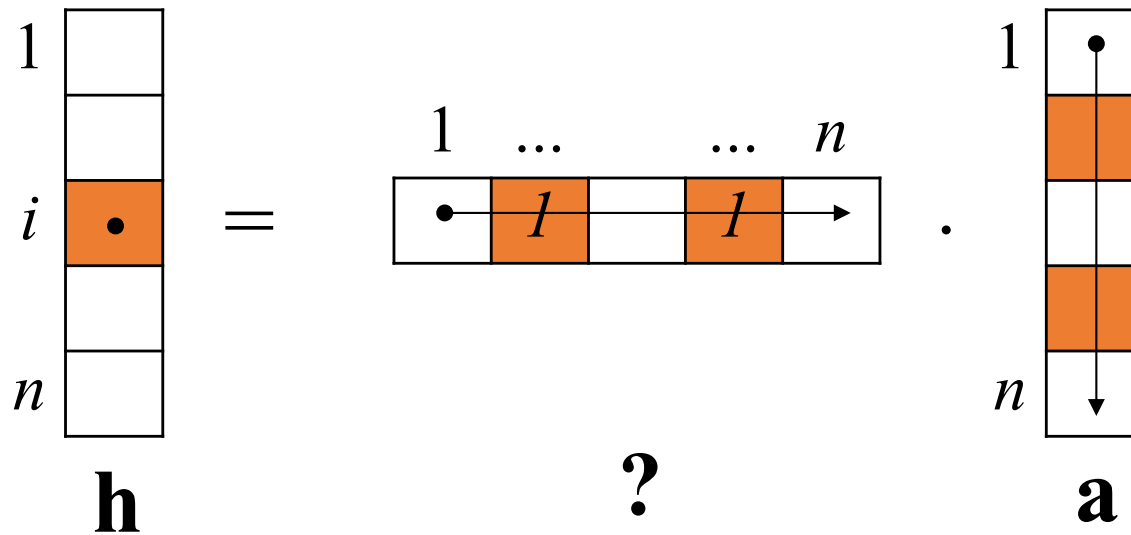
|              |                    |     |                    |     |                    |
|--------------|--------------------|-----|--------------------|-----|--------------------|
| • <b>h</b> = | <b>h</b> ( $u_0$ ) | ... | <b>h</b> ( $u_i$ ) | ... | <b>h</b> ( $u_n$ ) |
| • <b>a</b> = | <b>a</b> ( $u_0$ ) | ... | <b>a</b> ( $u_i$ ) | ... | <b>a</b> ( $u_n$ ) |

$$\mathbf{h}(x) = \sum_{x \rightarrow y} \mathbf{a}(y)$$

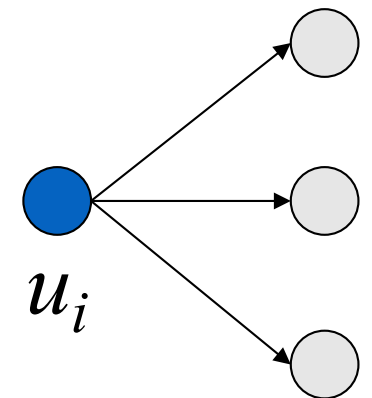
- To compute **h** i-th component we take all **a** components corresponding to  $u_i$  outpages and sum them
  - That is, sum all **a** components, multiplying by 1 those of the outpages and by 0 the other ones



# Matrices and vectors (2/4)

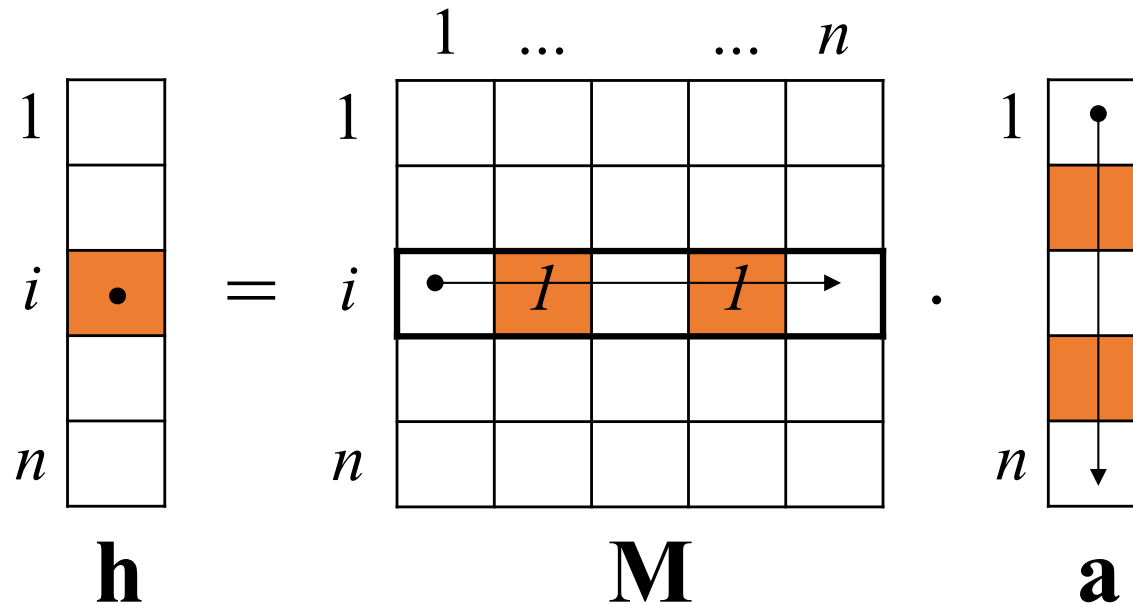


$$\mathbf{h}(x) = \sum_{x \rightarrow y} \mathbf{a}(y)$$

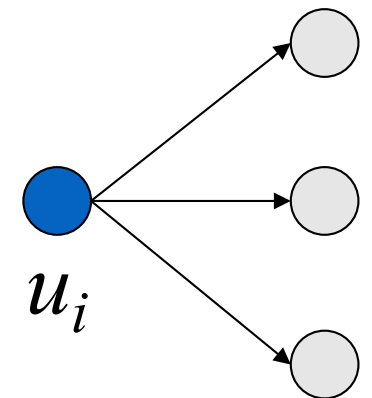


- That is, multiply a row vector by  $\mathbf{a}$
- But which is the (row) vector that has 1s for the pages linked from page  $i$  and 0s otherwise?
- The  $i$ -th row of the adjacency matrix!

# Matrices and vectors (3/4)



$$\mathbf{h}(x) = \sum_{x \rightarrow y} \mathbf{a}(y)$$



- Thus, multiplying the  $i$ -th row of  $\mathbf{M}$  by  $\mathbf{a}$  we get the  $i$ -th component of  $\mathbf{h}$ ...
- ...and multiplying all  $\mathbf{M}$  by  $\mathbf{a}$  we get all  $\mathbf{h}$ !
- $\mathbf{h} = \mathbf{M}\mathbf{a}$

# Matrices and vectors (4/4)

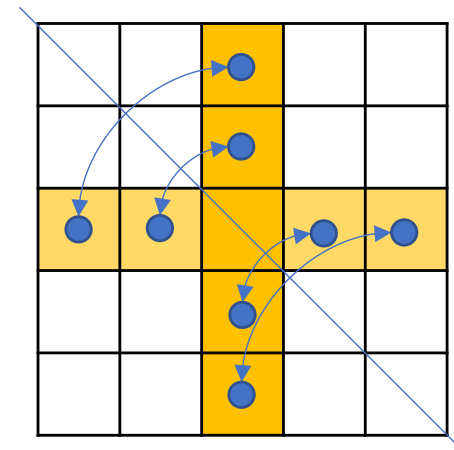
- Therefore

- $\mathbf{h} = \mathbf{M}\mathbf{a}$

- With a similar (exercise!) argument we get

- $\mathbf{a} = \mathbf{M}^T\mathbf{h}$

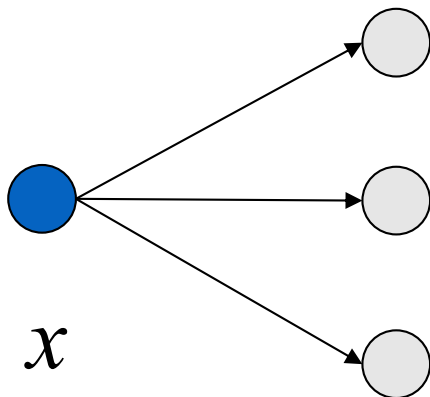
- ( $\mathbf{M}^T$  is the transpose of  $\mathbf{M}$ )



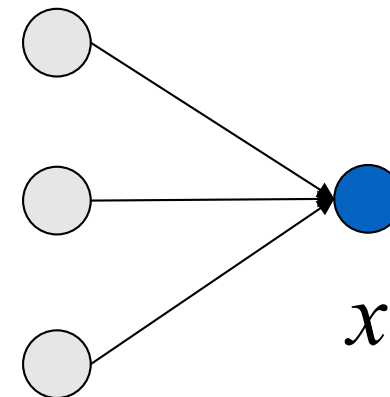
$\forall x \in BS$  compute  $\mathbf{h}(x)$  and  $\mathbf{a}(x)$

$$\mathbf{h}(x) = \sum_{x \rightarrow y} \mathbf{a}(y)$$

$$\mathbf{a}(x) = \sum_{y \rightarrow x} \mathbf{h}(y)$$



- The  $\mathbf{h}$  value of  $x$  depends on how  $x$  links high  $\mathbf{a}$  pages
- $\mathbf{h} = \mathbf{M}\mathbf{a}$

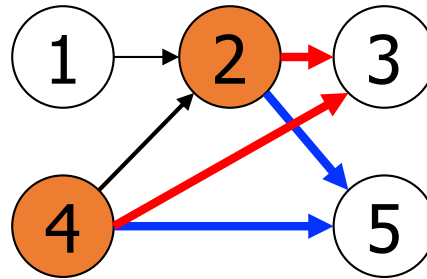


- The  $\mathbf{a}$  value of  $x$  depends on how  $x$  is linked from high  $\mathbf{h}$  pages
- $\mathbf{a} = \mathbf{M}^T \mathbf{h}$

# Eigenvectors

- We have:
  - $\mathbf{h} = \mathbf{M}\mathbf{a}$
  - $\mathbf{a} = \mathbf{M}^T\mathbf{h}$
- By substitution we get:
  - $\mathbf{h} = \mathbf{M}\mathbf{a} = \mathbf{M}\mathbf{M}^T\mathbf{h}$
  - $\mathbf{a} = \mathbf{M}^T\mathbf{h} = \mathbf{M}^T\mathbf{M}\mathbf{a}$
- $\mathbf{h}$  is the eigenvector of  $\mathbf{M}\mathbf{M}^T$  and  $\mathbf{a}$  is the eigenvector of  $\mathbf{M}^T\mathbf{M}$
- But what are  $\mathbf{M}\mathbf{M}^T$  and  $\mathbf{M}^T\mathbf{M}$ ?

# What is $\mathbf{MM}^T$ ? (1/2)



|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   | 1 |   |   |   |
| 2 |   |   | 1 |   | 1 |
| 3 |   |   |   |   |   |
| 4 |   | 1 | 1 |   | 1 |
| 5 |   |   |   |   |   |

$\mathbf{M}$

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   |   |   |   |   |
| 2 | 1 |   |   | 1 |   |
| 3 |   | 1 |   | 1 |   |
| 4 |   |   |   |   |   |
| 5 |   | 1 |   | 1 |   |

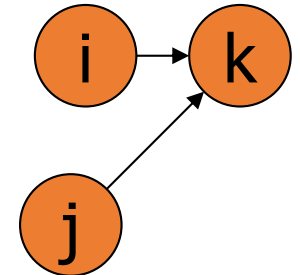
$\mathbf{M}^T$

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 1 |   |   | 1 |   |
| 2 |   | 2 |   | 2 |   |
| 3 |   |   |   |   |   |
| 4 | 1 | 2 |   | 3 |   |
| 5 |   |   |   |   |   |

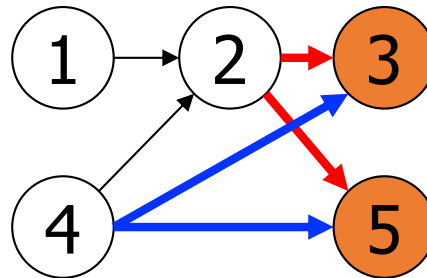
$\mathbf{MM}^T$

# What is $\mathbf{MM}^T$ ? (2/2)

- $\mathbf{MM}^T[i,j]$  is incremented by 1 when  $\exists k$  s.t.  $\mathbf{M}[i,k] = \mathbf{M}^T[k,j] = 1$ 
  - In the example, for  $\mathbf{MM}^T[4,2]$ ,  $k=3, 5$  ( $i = 4, j = 2$ )
  - But  $\mathbf{M}^T[k,j] = 1$  iff  $\mathbf{M}[j,k] = 1$
  - $i$  links  $k$  and  $j$  links  $k$ 
    - 4 links 3 and 2 links 3
    - 4 links 5 and 2 links 5
- $\mathbf{MM}^T[i,j]$  counts the nodes that are **co-linked** by  $i$  and  $j$ : the "**bibliographic couplings matrix**"
- (BTW,  $\mathbf{MM}^T$  is symmetric,  $\mathbf{MM}^T[i,j] = \mathbf{MM}^T[j,i]$  )



# What is $\mathbf{M}^T\mathbf{M}$ ? (1/2)



|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   |   |   |   |   |
| 2 | 1 |   |   | 1 |   |
| 3 |   | 1 |   | 1 |   |
| 4 |   |   |   |   |   |
| 5 |   | 1 |   | 1 |   |

$\mathbf{M}^T$

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   | 1 |   |   |   |
| 2 |   |   | 1 |   | 1 |
| 3 |   |   |   |   |   |
| 4 |   | 1 | 1 |   | 1 |
| 5 |   |   |   |   |   |

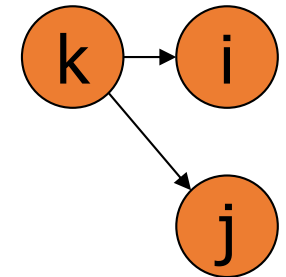
$\mathbf{M}$

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   |   |   |   |   |
| 2 |   | 2 | 1 |   | 1 |
| 3 |   | 1 | 2 |   | 2 |
| 4 |   |   |   |   |   |
| 5 |   | 1 | 2 |   | 2 |

$\mathbf{M}^T\mathbf{M}$

# What is $\mathbf{M}^T\mathbf{M}$ ? (2/2)

- $\mathbf{M}^T\mathbf{M}[i,j]$  is incremented by 1 when  $\exists k$  s.t.  $\mathbf{M}^T[i,k] = \mathbf{M}[k,j] = 1$ 
  - In the example, for  $\mathbf{M}^T\mathbf{M}[3,5]$ ,  $k=2, 4$  ( $i = 3, j = 5$ )
  - But  $\mathbf{M}^T[i,k] = 1$  iff  $\mathbf{M}[k,i] = 1$
  - $k$  links  $i$  and  $k$  links  $j$ 
    - 2 links 3 and 2 links 5
    - 4 links 3 and 4 links 5
    - If I add  $\mathbf{M}[1,5] = 1$  (link  $1 \rightarrow 5$ ) nothing changes; if I add also  $\mathbf{M}[1,3] = 1$  (link  $1 \rightarrow 3$ ), it changes!
- $\mathbf{M}^T\mathbf{M}[i,j]$  counts the nodes that **co-link**  $i$  and  $j$ , the co-citations  $(i, j)$ : it is the **co-citation matrix**
- (BTW,  $\mathbf{M}^T\mathbf{M}$  is symmetric,  $\mathbf{M}^T\mathbf{M}[i,j] = \mathbf{M}^T\mathbf{M}[j,i]$  )

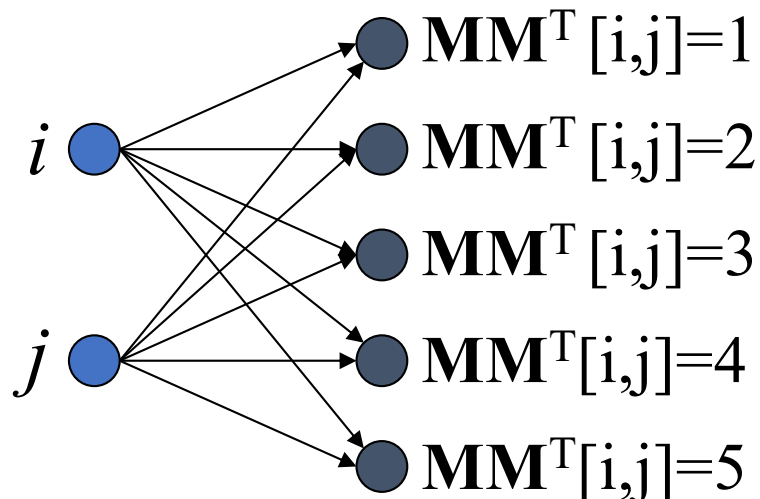


# Summary

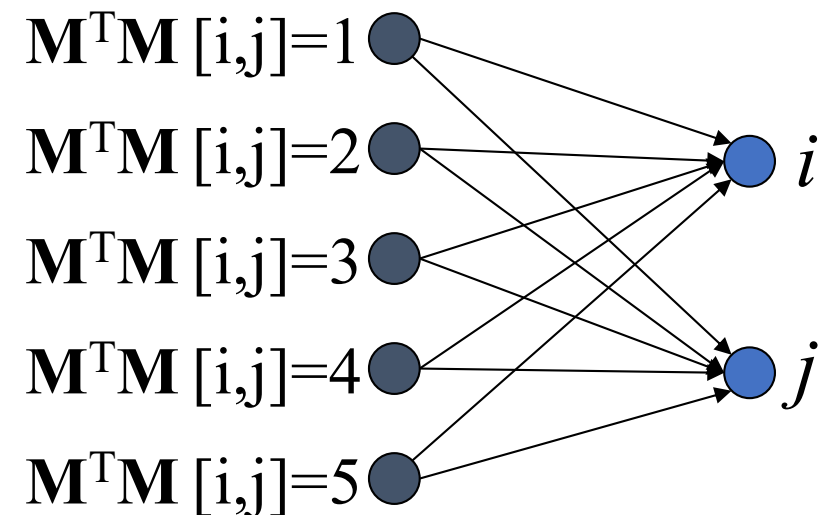
- Eigenvectors

- $\mathbf{h} = \mathbf{M}\mathbf{a}$ ;  $\mathbf{a} = \mathbf{M}^T\mathbf{h}$
- By substitution:  $\mathbf{h} = \mathbf{M}\mathbf{a} = \mathbf{M}\mathbf{M}^T\mathbf{h}$ ;  $\mathbf{a} = \mathbf{M}^T\mathbf{h} = \mathbf{M}^T\mathbf{M}\mathbf{a}$
- $\mathbf{h}$ : eigenvector of bibliographic coupling matrix, outlink based ( $\mathbf{M}\mathbf{M}^T$ )
- $\mathbf{a}$ : eigenvector of co-citations matrix, inlink based ( $\mathbf{M}^T\mathbf{M}$ )

Bibliographic coupling: how  $i$  and  $j$  cite the same docs.



Co-citations: how  $i$  and  $j$  are cited in the same docs.



# Comparisons and remarks

- PageRank
  - **Pros:** not spammable, quality index for the whole Web, used in Google (and in all SEs)
  - **Contra:** not query specific, ok for large graphs, complex computation
- HITS, Kleinberg
  - **Pros:** query specific, works on small graphs, hubs and authorities
  - **Contra:** easily spammable (on the hubs), complex computation (but on small graphs...), convergence?

# Prestige and Centrality

- Well known notions in Social Network Analysis and Bibliometrics
- Prestige ("status") in Social networks:
  - «we are involved in an "infinite regress": [an actor's status] is a function of the status of those who choose him; and their [status] is a function of those who choose them, and so on ad infinitum»
  - [Seeley, 1949] (!)
- Does it ring a bell? ;-)
- Eigenvectors, fixed points, power iteration  
 $\mathbf{r}_{i+1} = \mathbf{r}_i \cdot \mathbf{P}$ , ... well known before 1970
- Centrality: how much a node is "in the center" of a network

# Biblio

- [Brin & Page 98] The Anatomy of a Large-Scale Hypertextual Web Search Engine, WWW7, 107–117, 1998, <http://www-db.stanford.edu/~backrub/google.html>
- [Page et al. 98] L. Page, S. Brin, R. Motwani, T. Winograd. The PageRank Citation Ranking: Bringing Order to the Web. Manoscritto non pubblicato, <http://ilpubs.stanford.edu:8090/422/1/1999-66.pdf>  
<http://wwwdb.stanford.edu/~backrub/pageranksub.ps>
- [Kleinberg 99] J. Kleinberg, Authoritative sources in a hyperlinked environment, Journal of the ACM, 1999

# From the Web to Social Media...

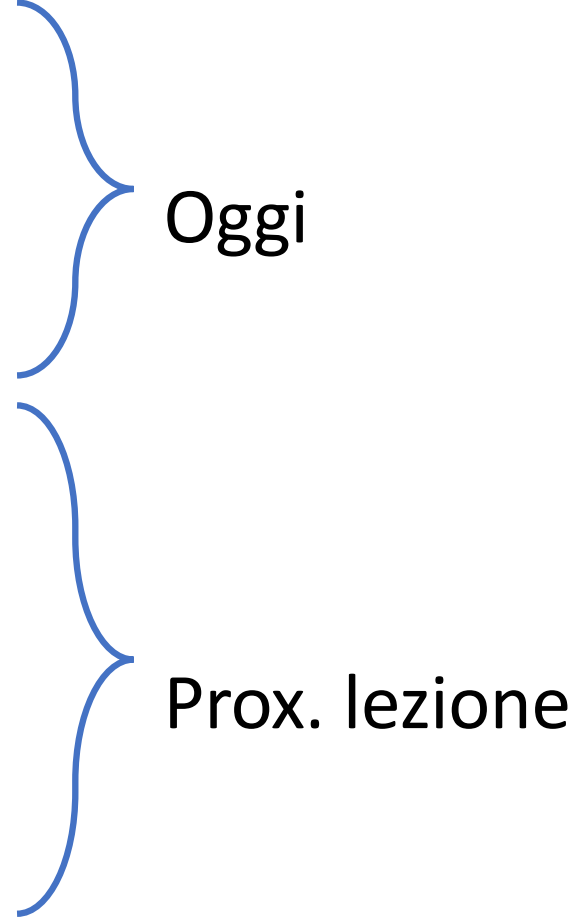
- Facile
- Da centralità di pagine Web a
- Centralità di Social Media users / profiles
  - Facebook – indiretta
  - Twitter – diretta
- Ma anche di post / condivisioni di post / ...
- Es.: HITS su profili Twitter --> hub e authority...

# Riassunto

- Centralità
  - Degree
  - Betweenness
  - Closeness
  - (Ricorsiva) PageRank e HITS

# Scaletta

- Centralità
  - Degree
  - Betweenness
  - Closeness
  - (Ricorsiva) PageRank e HITS
- Social Network Analysis (SNA)
- Transitività e Reciprocità
  - E coefficiente di clustering
- Bilanciamento
- Distribuzione dei gradi
- Distanza media / piccolo mondo

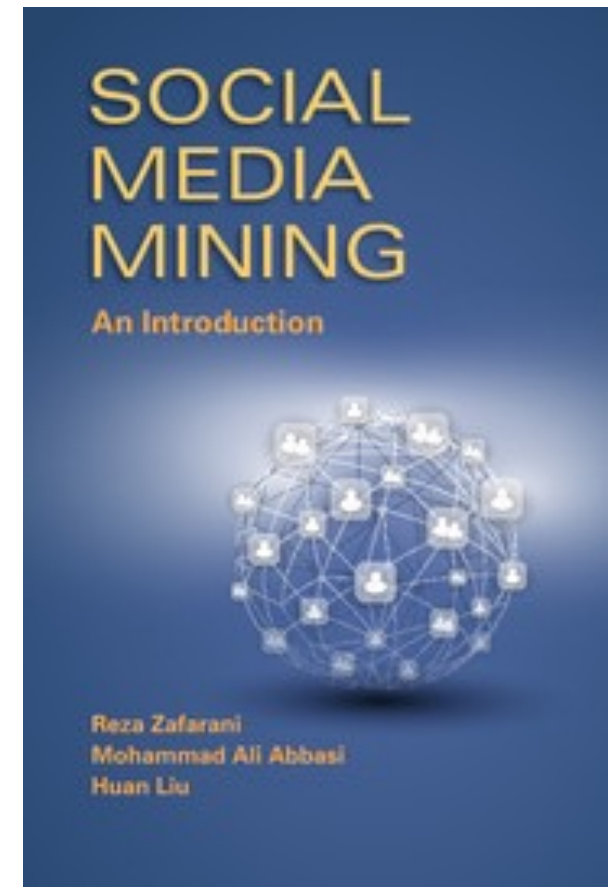


Oggi

Prox. lezione

# Biblio

- Reza Zafarani, Mohammad Ali Abbasi, Huan Liu.  
*Social Media Mining - An Introduction*, Cambridge University Press, 2014
- <http://socialmediamining.info/>
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# Ma anche, per approfondire

- D. Easley and J. Kleinberg. *Networks, Crowds, and Markets: Reasoning About a Highly Connected World*. Cambridge University Press, 2010.
- M. Newman. *Networks. Second Edition*. Oxford University Press, 2018. [oggi: Cap. 7]
- F. Menczer, S. Fortunato, C.A. Davis. *A First Course in Network Science*. Cambridge University Press, 2020.
- A.L. Barabási. *Network Science*. Cambridge University Press, 2016
- Stanley Wasserman e Katherine Faust. *Social Network Analysis: Methods and Applications*. Cambridge University Press. 1994
- Articoli scientifici

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R. Zafarani, M. A. Abbasi, and H. Liu, *Social Media Mining: An Introduction*, Cambridge University Press, 2014.  
Free book and slides at **<http://socialmediamining.info/>**

or include a link to the website:  
**<http://socialmediamining.info/>**